

Aldosterone and Mineralocorticoid Receptors: Renal and Extrarenal Roles

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KEY POINTS

- Inappropriate aldosterone hypersecretion relative to sodium status is much more common in people with hypertension than has been generally appreciated and is even found in a significant percentage of people with normal blood pressure (~10%).
- Most unilateral aldosterone-producing adenomas harbor disease-causing gene mutations, the most common of which is in the K⁺ channel gene, *KCNJ5*. Germline mutations in *KCNJ5* cause familial bilateral adrenal hyperplasia. Most recently, the chloride channel gene, *CLCN2*, has been implicated in early-onset primary aldosteronism, with clinical findings most consistent with bilateral hyperplasia.
- Evidence supports the idea that the actions of aldosterone on the distal convoluted tubule are indirect and mediated by changes in K⁺. Aldosterone stimulates the epithelial sodium channel (ENaC) in the connecting tubule and cortical collecting ducts, and the ensuing hypokalemia acts directly in distal convoluted tubule cells to stimulate the sodium-chloride cotransporter, NCC.
- Serum/glucocorticoid regulated kinase 1 (SGK1) gene transcription is regulated by aldosterone, and it in turn stimulates ENaC, thereby enhancing Na⁺ reabsorption and K⁺ secretion. SGK1 also responds to and integrates a variety of other hormonal and nonhormonal signals, including insulin and IL-17, which stimulate SGK1 activity and contribute to salt-sensitive hypertension.
- Studies in mice demonstrate a role for T cells in the pathogenesis of hypertension through a mechanism involving IL-17A and SGK1. Mice lacking T cell SGK1 are protected from angiotensin II-induced hypertension and renal inflammation.

In mammals, the control of extracellular fluid volume and blood pressure is intimately intertwined with the regulation of epithelial ion transport. Aldosterone, which is essential for survival, is the central hormone regulating the relevant epithelial transport processes, particularly of ions such as Na^+ , K^+ , and Cl^- . All circulating aldosterone is generated in the adrenal glomerulosa, where its synthesis and secretion are primarily under the control of angiotensin II and potassium, and its major epithelial actions occur in the distal nephron and colon. The former extends from the late distal convoluted tubule (DCT) through the connecting segment and the entire cortical and medullary collecting ducts. These segments, rich in mineralocorticoid receptors (MRs), are often referred to as the “aldosterone-sensitive distal nephron” (ASDN).¹ Most, if not all, effects of aldosterone are mediated by the MR, a hormone-regulated transcription factor related closely to the glucocorticoid receptor and more distantly to other members of the nuclear receptor superfamily. The physiologic effects of aldosterone on epithelia entail direct gene-regulatory actions of the MR. Thus a sound foundation for understanding aldosterone’s physiologic effects on the extracellular fluid, blood pressure, and electrolyte concentrations can be understood through familiarity with the MR-dependent effects on the transcription of various genes, which, in turn, alter epithelial ion transport. Aldosterone actions in certain disease states involve both genomic and nongenomic effects in epithelial and nonepithelial tissues. Furthermore, physiologic and pathophysiologic effects of cortisol are also mediated in part by the MR, which binds cortisol with high affinity. This chapter addresses the cellular and molecular mechanisms underlying aldosterone—and, to some extent, cortisol—action, focusing primarily on effects on ion transport in epithelia but also highlighting key aspects of nonepithelial actions, which are of substantial importance to its pathophysiologic effects.

GENERAL INTRODUCTION TO ALDOSTERONE AND MINERALOCORTICOID RECEPTORS

Steroid hormones are derived from cholesterol and produced in systemically relevant amounts in a relatively

narrow range of tissues (e.g., adrenal glands, gonads, placenta, and skin). In mammalian physiology, six classes of steroid hormones are commonly recognized—mineralocorticoid, glucocorticoid, androgen, estrogen, progestin, and the secosteroid vitamin D_3 . This classification was based on observed effects of these hormones and has proven robust, despite current appreciation of a much more diverse physiology of steroid hormones over and above their classic roles. In further support of this original classification is the characterization of six intracellular receptors—mineralocorticoid, MR; glucocorticoid, GR; androgen, AR; estrogen, ER; progestin, PR; and vitamin D_3 , VDR. As further addressed later, it is now appreciated that a one-to-one relationship between receptor and hormone does not hold, and this is particularly the case for the MR.

Aldosterone was isolated and characterized in 1953. Crucial for its isolation was the application of radioisotopic techniques to measure $[\text{Na}^+]$ and $[\text{K}^+]$ flux across epithelia in the laboratory of Sylvia Simpson, a biologist, and Jim Tait, a physicist.^{2,3} Because of this, the active principle was initially called “electrocortin”; the name was soon changed to aldosterone when its unique aldehyde (rather than methyl) group at carbon 18 was discovered in collaborative studies between investigators in London and Basel.³ Aldosterone is commonly depicted so as to highlight this aldehyde group (Fig. 12.1, right). In vivo, the very reactive aldehyde group cyclizes with the β -hydroxyl group at carbon 11 to form the 11,18- hemiacetal and, in addition, may exist in an 11,18-hemiketal form. This cyclization of the 11 β -hydroxyl group protects aldosterone from dehydrogenation by the enzyme 11 β -hydroxysteroid dehydrogenase in epithelial tissue and by some neuronal and smooth muscle cells, which enables it to activate the epithelial MR and thus regulate ion transport at very low (subnanomolar) circulating levels.^{4,5}

There is broad evidence that aldosterone is not the only cognate ligand for the MR, its essential effects via the MR on epithelial ion transport notwithstanding. The MR is found in high abundance in the hippocampus and cardiomyocytes and, in these nonepithelial tissues—which lack 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2; see later, “11 β -Hydroxysteroid Dehydrogenase Type 2”)—they are essentially constitutively occupied by glucocorticoids (cortisol in humans, corticosterone in rodents). This is due

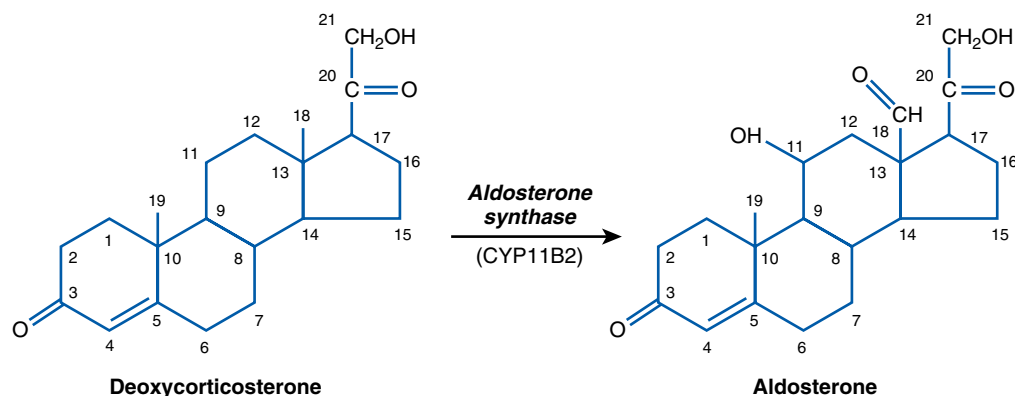


Fig. 12.1 Final step in aldosterone synthesis. Note that the aldehyde form of aldosterone is shown. Most aldosterone (>99%) exists as the hemiacetal form, which is cyclized and does not allow access of 11 β -hydroxysteroid dehydrogenase type 2 (11 β -HSD2) to the 11-hydroxyl. See text for details.

to the comparable affinity and markedly higher plasma-free levels (≥ 1000 -fold) of endogenous glucocorticoids compared with those of aldosterone. In terms of evolution, the MR appeared well before aldosterone synthase (e.g., in fish).⁶ It was commonly assumed that the MR and glucocorticoid receptors share a common immediate evolutionary precursor,⁷ although this has been challenged on sequence grounds,⁸ which implicate the MR as the first of the MR-GR-AR-PR subfamily to branch off an ancestral receptor. A final reason not to equate the MR and aldosterone action derives from a comparison of the mineralocorticoid receptor knockout and aldosterone synthase knockout ($AS^{-/-}$) phenotypes. MR knockout mice (which lack all functional MRs) cannot survive sodium restriction; $AS^{-/-}$ mice (which have no detectable aldosterone) survive even stringent sodium restriction but die when their fluid intake is restricted to that of wild-type animals.⁹ The survival of $AS^{-/-}$ mice on a low- Na^+ intake may reflect, in part, Na^+ retention via renal tubular intercalated cells, in which MRs (not 11β -HSD2-protected) are activated by glucocorticoids in the context of high ambient angiotensin concentrations.^{10,11} Their inability to survive fluid restriction suggests an as yet poorly defined dependence on aldosterone for vasopressin action.¹² Potassium homeostasis is also surprisingly intact in $AS^{-/-}$ mice, although they do not tolerate extremes of K^+ loading.¹³

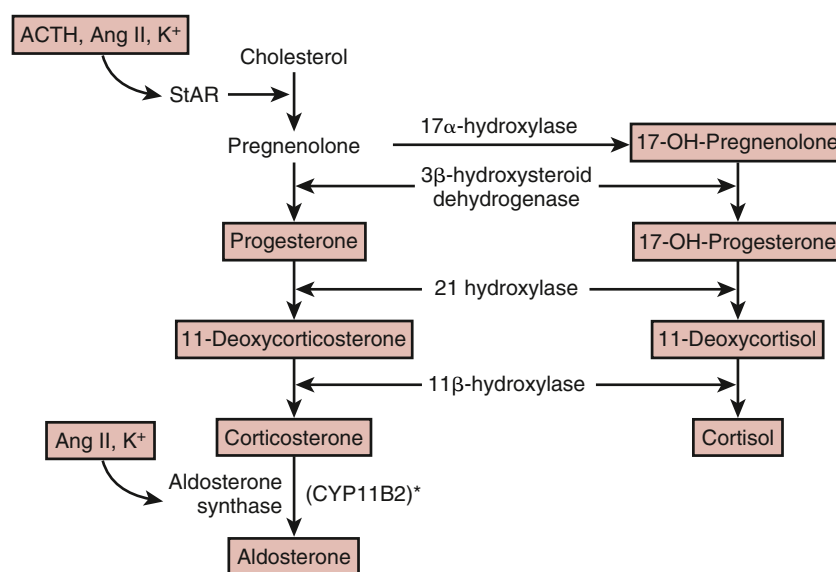
ALDOSTERONE SYNTHESIS

Aldosterone is synthesized in the adrenal cortex, which has three functional zones. The outermost layer of cells represents the zona glomerulosa, which is the unique site of aldosterone biosynthesis in normal physiology (see later;

aldosterone is produced in excessive amounts in patients with glucocorticoid-remediable aldosteronism). Cortisol is synthesized in the middle zone, the zona fasciculata, and the innermost zona reticularis secretes adrenal androgens in many species, including humans, but not in rats or mice. Normally, the glomerulosa secretes aldosterone at the rate of 50 to 200 $\mu\text{g}/\text{day}$ to give plasma levels of 4 to 21 $\mu\text{g}/\text{dL}$; in contrast, secretion of cortisol is at levels 200- to 500-fold higher. Underlying the separate synthesis of cortisol and aldosterone is expression of the enzyme 17α -hydroxylase uniquely in the zona fasciculata and that of aldosterone synthase uniquely in the glomerulosa.

In most species, aldosterone synthase, or cytochrome P450 (CYP) enzyme 11β 2, is responsible for the conversion of deoxycorticosterone to aldosterone in a three-step process of sequential 11β -hydroxylation, 18 -hydroxylation, and 18 -methyl oxidation, to produce the characteristic C18-aldehyde from which aldosterone derives its name (see Fig. 12.1, left). Although CYP11B2 is distinct from CYP11B1 (11β -hydroxylase) in most species,^{14,15} in some species (e.g., bovine), only a single CYP11B is expressed. How this enzyme is responsible for the three-step process of aldosterone synthesis in the glomerulosa but not the fasciculata has yet to be determined.

Fig. 12.2 also illustrates key steps in the biosynthesis of cortisol to illustrate the overlap and similarities with those of aldosterone. The genes encoding CYP11B1 and CYP11B2 lie close to one another on human chromosome band 8q24.3, so an unequal crossing over at meiosis has been shown to be responsible for the syndrome of glucocorticoid-remediable aldosteronism (now known as “familial hyperaldosteronism type I”), in which the 5' end of the CYP11B1 is fused to the 3' end of CYP11B2. The chimeric gene product¹⁶ is expressed



*Aldosterone synthase = CYP11B2

Fig. 12.2 Overview of aldosterone synthetic pathway showing key regulatory nodes. Note that adrenocorticotrophic hormone (*ACTH*), angiotensin II (*Ang II*), and K^+ regulate steroidogenic acute regulatory protein (*StAR*), which stimulates cholesterol uptake by mitochondria and thus substrate availability for synthesis of all of the steroid hormones. Aldosterone synthase (gene name *CYP11B2*), which is selectively expressed in the adrenal glomerulosa, mediates the final step in aldosterone synthesis. It is also regulated by *Ang II* and K^+ . Aldosterone synthesis is shown (left). Cortisol synthesis is also shown (right) to emphasize the interconnections and similarities between these pathways.

in the fasciculata and responds to adrenocorticotrophic hormone (ACTH) with aldosterone synthesis, producing a syndrome of juvenile-onset hyperaldosteronism and hypertension.

Normal glomerulosa secretion of aldosterone is primarily regulated by angiotensin II in response to posture and acute lowering of circulating volume, to plasma $[K^+]$ in response to elevated potassium levels, particularly in settings of Na^+ deficiency,¹⁷ and to ACTH to the extent of entrainment of the circadian fluctuation in plasma aldosterone levels with those of cortisol. Aldosterone secretion is lowered by high levels of atrial natriuretic peptide and by the administration of heparin, somatostatin, and dopamine. As yet, incompletely characterized molecules of adipocyte origin have been shown to stimulate aldosterone secretion in vitro, and roles in metabolic syndrome have been proposed on this basis.¹⁸

Angiotensin and plasma $[K^+]$ stimulate aldosterone secretion primarily by increasing the expression and activity of key steroidogenic enzymes, as well as the steroidogenic acute regulatory protein (StAR).¹⁹ StAR is required for cholesterol transport into mitochondria and hence for its availability for steroid synthesis.²⁰ Regulated steroidogenic enzymes include side chain cleavage enzyme 3β -hydroxysteroid dehydrogenase and, most notably, aldosterone synthase. Common to the mechanism of stimulation by angiotensin II and $[K^+]$ is elevation of intracellular $[Ca^{2+}]$. Angiotensin II activates the G-protein-coupled angiotensin type I receptor (AT1R) in the glomerulosa cell membrane, which in turn activates phospholipase C, which catalyzes the hydrolysis of phosphatidylinositol 4,5-bisphosphate (PIP2) to inositol trisphosphate (IP3) and diacylglycerol (DAG). DAG stimulates protein kinase C, whereas IP3 stimulates Ca^{2+} release from intracellular stores, both of which affect the aldosterone biosynthetic pathway. AT1R also separately stimulates Ca^{2+} influx, which is important for sustained stimulation of aldosterone secretion.²¹ Elevated $[K^+]$ increases intracellular $[Ca^{2+}]$ by depolarizing the cell membrane and activating voltage-sensitive Ca^{2+} channels.^{22,23} Patients taking angiotensin-converting enzyme inhibitors or angiotensin receptor blockers usually show a degree of suppression of aldosterone secretion, reflected in a modest (0.2–0.3 mEq/L) elevation in plasma $[K^+]$ levels. This is often sufficient to establish a new steady state, with plasma aldosterone levels rising into the normal range, a process best termed “breakthrough” rather than “escape,” given the time-honored usage of the latter to refer to the escape from progression of the salt and water effect of mineralocorticoid excess in the medium and long term.²⁴

Studies have shed some light on the regulation of aldosterone production by the adrenal glomerulosa in health and disease. Choi and associates found recurrent somatic mutations in the K^+ channel Kir3.4 (encoded by the gene *KCNJ5*), which were present in more than one-third of spontaneous human aldosterone-producing adenomas studied.²⁵ These mutations increased Na^+ conductance through Kir3.4 and resulted in increased Ca^{2+} entry and enhanced aldosterone production and glomerulosa cell proliferation. Interestingly, an inherited mutation in *KCNJ5* is associated with hypertension associated with marked bilateral adrenal hyperplasia (now known as “familial hyperaldosteronism type III” [FH-III]).²⁶ These findings suggest that *KCNJ5* may provide tonic inhibition of

aldosterone production and glomerulosa cell proliferation. In glomerulosa cells harboring the mutant channel, both proliferation rate and aldosterone synthesis are increased. These initial studies were continued and extended by a much wider survey by Boulkroun and colleagues²⁷; subsequently, less common but similarly somatic mutations in genes expressed in the adrenal cortex have been associated with hyperaldosteronism caused by adrenal adenomas^{28,29} (see later, under “Primary Aldosteronism”).

MECHANISMS OF MINERALOCORTICOID RECEPTOR FUNCTION AND GENE REGULATION

Mammals cannot survive without the MR, except with substantial NaCl supplementation. This member of the nuclear receptor superfamily appears to have both genomic and nongenomic actions; however, the latter do not appear to play a significant role in the control of epithelial ion transport. This section thus focuses exclusively on the function of the MR as a hormone-regulated transcription factor.

GENERAL FEATURES AND SUBCELLULAR LOCALIZATION

In the presence of agonists, the MR binds to specific genomic sites and alters the transcription rate of a subset of genes. Fig. 12.3 shows the fundamental paradigm of MR function. All nuclear receptors shuttle in and out of the nucleus; however, in the absence of hormones, some, such as the estrogen and vitamin D receptors are predominantly nuclear, whereas others, like glucocorticoid receptors, are

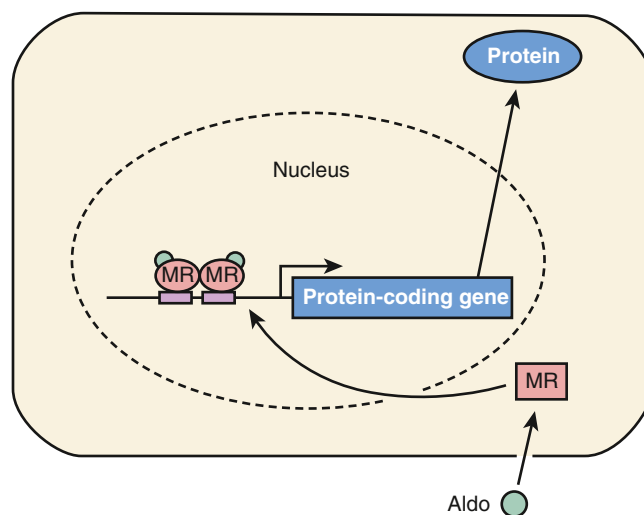


Fig. 12.3 General mechanism of aldosterone action through the mineralocorticoid receptor (MR). This simple schematic shows the general features of MR regulation of a “simple” hormone response element (HRE), common to a large subset (but not all) of aldosterone-stimulated genes. Note that in the absence of hormone, MR is found in both nucleus and cytoplasm (see Fig. 12.4). Aldosterone (Aldo) triggers nuclear translocation of cytoplasmic MR, binding as a dimer to HREs, and stimulation of transcription initiation complex formation (arrow, upstream of the so-called protein-coding gene, defines the site of transcription initiation). See text for further details.

almost exclusively cytoplasmic. In the absence of hormones, the MR is distributed relatively evenly between nuclear and cytoplasmic compartments, but in the presence of hormones, it is highly concentrated in the nucleus (Fig. 12.4).^{30,31} It is also notable that in addition to this marked change in MR cellular distribution, its subnuclear organization and protein–protein interactions are also changed.³¹ Like its close cousin the glucocorticoid receptor, the unliganded MR (in the absence of hormone) is complexed with a set of chaperone proteins, which include the heat shock proteins hsp90, hsp70, and hsp56 and immunophilins.^{32,33} This chaperone complex is essential for several aspects of MR function, notably high-affinity hormone binding and trafficking to the nucleus.³² It was thought for many years that after binding hormone, the hsp90-containing chaperone complex is jettisoned. However, it has become clear that this complex remains associated with the receptor and plays an important role in nuclear trafficking.³³ Several members of the immunophilin family, including FKBP52, FKBP51, and CyP40 are present in the chaperone complex

and provide a bridge between hsp90 and the cytoplasmic motor protein dynein, which moves the receptor-hsp90 complex retrogradely along microtubules to the nuclear envelope.³² Here, the receptor is handed off to the nuclear pore protein, importin- α , and translocated into the nucleus, where it functions as a transcription factor, stimulating the transcription of certain genes and repressing the transcriptional activity of others. In the regulation of ion transport, stimulation of key target genes is paramount. Transcriptional repression may be essential for effects in nonepithelial cells including neurons, cardiomyocytes, smooth muscle cells, and macrophages.³⁴

DOMAIN STRUCTURE OF MINERALOCORTICOID RECEPTORS

The MRs of all vertebrates are highly conserved and, like other steroid and nuclear receptors, can be divided into three major domains (Fig. 12.5):

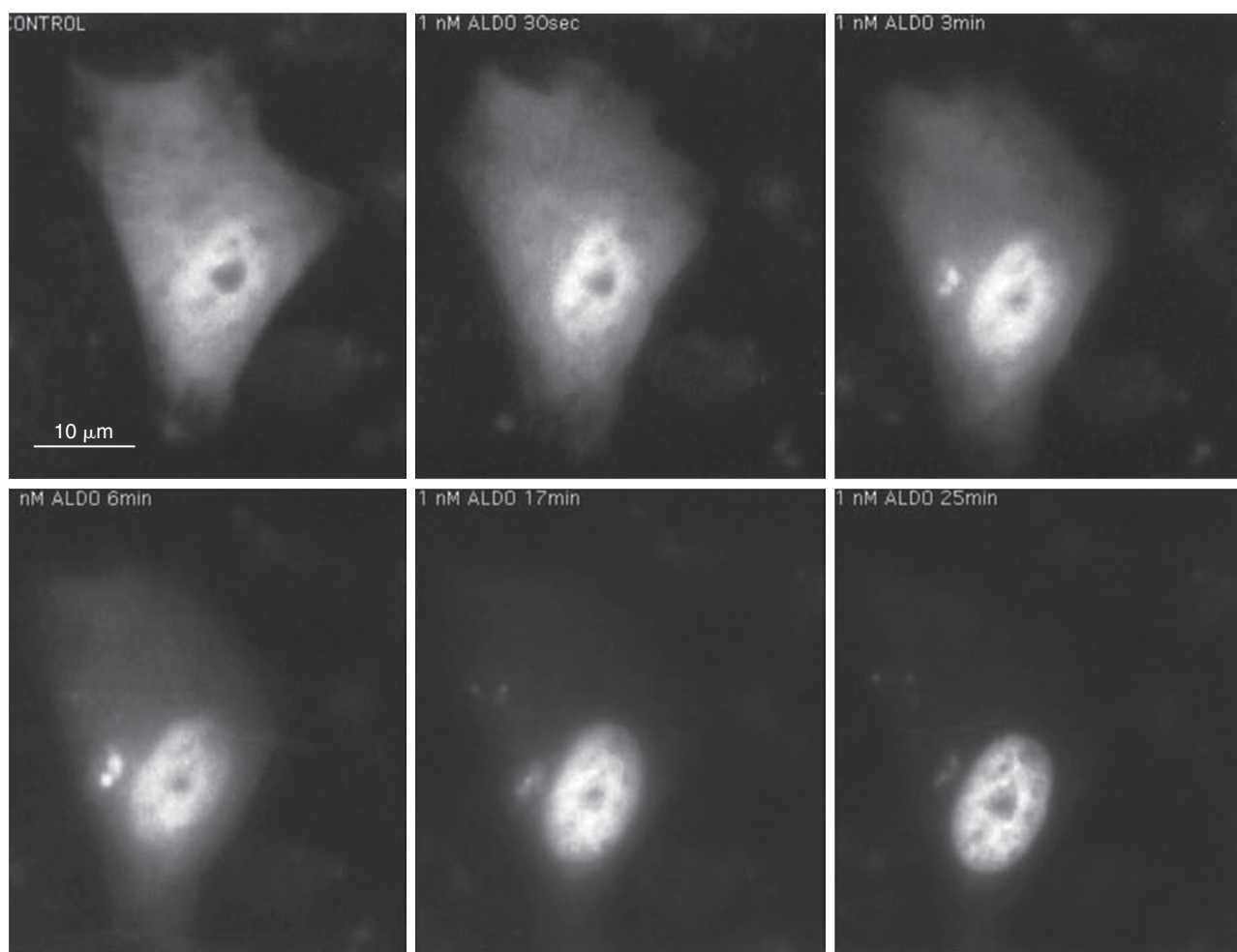


Fig. 12.4 Time-dependent nuclear translocation of the mineralocorticoid receptor (MR) in the presence of aldosterone. Cultured cells expressing green fluorescent protein–MR fusion protein (GFP–MR) were grown in a steroid-free medium and treated with 1 nmol/L aldosterone. Translocation of GFP–MR was followed in real time, and images were captured at indicated times. It is notable that the nuclear accumulation of GFP–MR started within 30 seconds, was half-maximal at 7.5 minutes, and was complete by 10 minutes. Control: MR distribution before addition of aldosterone. (From Fejes-Toth G, Pearce D, Naray-Fejes-Toth A. Subcellular localization of mineralocorticoid receptors in living cells: effects of receptor agonists and antagonists. *Proc Natl Acad Sci U. S. A.* 1998; 95[6]:2973–2978.)

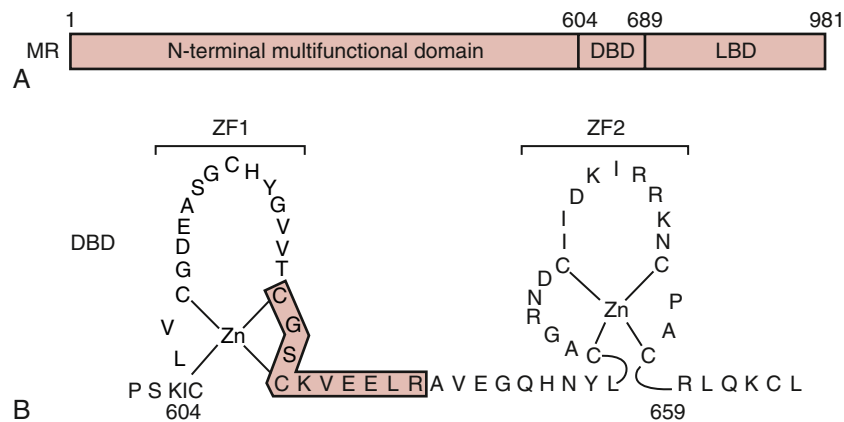


Fig. 12.5 Domain structure of the mineralocorticoid receptor (MR). The size and amino acid designations used here are for rat MR (981 amino acids total); they apply with minor variations to human MR (984 amino acids total). (A) Strip diagram of MR. The N-terminus is to the *left*, C-terminus to the *right*. (B) Schematic of MR DNA-binding domain (DBD), showing the two zinc fingers and the positions of the coordinating Zn ions. *Boxed region* is the α helix, which intercalates into the major groove of the DNA and provides the major protein–DNA interaction contacts. The dimerization interface comprises amino acids within the second zinc finger, which form van der Waals and salt bridge interactions. *LBD*, Ligand/hormone-binding domain.

1. An N-terminal transcriptional regulatory domain
2. A central DNA-binding domain (DBD)
3. A C-terminal ligand/hormone-binding domain (LBD)

Each of these broadly defined domains has more than one function, and not all the functions can be neatly assigned to separate distinct domains; however, much of the action of MRs can be understood from this point of view.

DNA-BINDING DOMAIN

The essential quality of MR function as a transcription factor is its ability to bind specifically to DNA. This protein–DNA interaction is mediated by the receptor’s compact modular DBD (amino acids 603–688 of human MR; see Fig. 12.5), which forms a variety of contacts with a specific 15-nucleotide DNA sequence termed a “hormone response element” (HRE). Receptor binding to the HRE in the vicinity of regulated genes promotes the recruitment of coactivators and components of the general transcription machinery, such as the TATA-binding protein, which binds to the thymidine- and adenosine-rich DNA sequence found upstream of many genes and is required for correct transcription initiation. These types of HREs have been identified near or in many of the key MR-regulated genes, such as serum- and glucocorticoid-regulated kinase 1 (*SGK1*), glucocorticoid-induced leucine zipper (*GILZ*), and amiloride-sensitive sodium channel subunit α (α -ENaC). Although, in many cases, differential binding to HREs is a key determinant of the specificity of many transcription factors, it should be noted that some steroid receptors (notably the MR and glucocorticoid receptor) have only minor (<10%) differences within this domain and have identical DNA-binding properties.³⁵ Specificity in these cases is determined through other mechanisms.^{34,36}

The canonical MR HRE is a 15-nucleotide sequence that forms a partial palindrome (inverted repeat), which binds a receptor homodimer. A dimer interface embedded within the DBD is essential for the MR to form these

requisite homodimers, as well as to form heterodimers with the glucocorticoid receptor.^{37,38} Mutations that disrupt this interface have complex effects on receptor activity in animals³⁹ and cultured cells,⁴⁰ and similar mutations in other receptors (the androgen receptor in particular) result in disease states.⁴¹ Also, in at least one kindred of the autosomal dominant form of pseudohypoaldosteronism type I (PHA I), the MR DBD mutation appears to be causative, although the mechanistic basis has not been elucidated.⁴²

In addition to supporting DNA binding and dimerization, the DBD also harbors a nuclear localization signal,^{43,44} as well as surfaces that contact distant parts of the receptor and that mediate interactions with other proteins, as has been shown for the glucocorticoid receptor and, in some cases, the MR.^{45–47}

LIGAND/HORMONE-BINDING DOMAIN

The MR LBD comprises amino acids 689 to 981 (see Fig. 12.5A). Like the DBD, the LBD has multiple functions; in addition to binding with high affinity to various MR agonists and antagonists, it also harbors interaction surfaces for coactivators, dimerization, and N-C interactions.⁴⁸ The MR is distinct from the glucocorticoid receptor in that it binds with equally high affinity to cortisol, corticosterone (the physiologic glucocorticoid in rats and mice), and aldosterone. Indeed, as discussed later, the MR appears to function as a high-affinity glucocorticoid receptor in some tissues including the brain and heart.

High-resolution representations of the crystal structures of wild-type and mutant MRs have identified the structural features of the LBD and specific amino acid contacts involved in binding to the mineralocorticoid desoxycorticosterone.^{49,50} Key features include the following:

1. The LBD of the MR, like that of other nuclear receptors, is arranged into 11 α -helices and four small β -strands.
2. The C-terminal alpha-helix (α -helix), H12, contains the activation function AF2.

3. Ligand is deeply embedded into a pocket comprising α -helices H3, H4, H5, H7, and H10 and two β strands (β -strands); numerous contacts are made between amino acids of the pocket and hormone.

This accounts for the slow off-rate and high affinity of aldosterone, corticosterone, and cortisol for the MR.⁵⁰ The crystal structure of the mutant (S810L) MR, in which progesterone acts as an agonist rather than an antagonist,⁵¹ reveals that H12 is stabilized with AF2 in the active conformation.⁴⁹ The crystal structure of wild-type MR LBD also provides insight into the mechanisms underlying some forms of PHA I. Notably, MR/S810L has an LBD mutation in helix 5, which is predicted to disrupt interaction with the steroid ring structure,⁵² whereas Q776R and L979P have been demonstrated to have markedly reduced aldosterone binding.⁴² Structural analysis reveals that Q776 is located in helix H3 at the extremity of the hydrophobic ligand-binding pocket and anchors the steroid C3-ketone group.

MR binds cortisol and corticosterone with an affinity similar to that of aldosterone. 11 β -HSD2 is an essential determinant of aldosterone specificity, through its effect to metabolize glucocorticoids to their receptor-inactive keto-congeners in collecting duct principal cells, as discussed later. In tissues that do not coexpress 11 β -HSD2, the physiologic ligand for MR is cortisol (corticosterone in rats and mice).⁵³⁻⁵⁵ The extent to which such “unprotected” MR can be pathophysiologically activated in aldosterone excess states is discussed later (see “Disease States: Primary Aldosteronism” and “Nonepithelial Actions of Aldosterone”).

N-TERMINAL DOMAIN

As its noncommittal name implies, the N-terminal region of the MR has diverse functions, which appear to revolve primarily around protein–protein interactions and the recruitment of coactivators and corepressors. It has two potent transcriptional regulatory motifs, usually termed *AF1a* and *AF1b*.^{48,56} This domain bears some functional and sequence similarity to the homologous region of the glucocorticoid receptor and is capable of stimulating gene transcription when fused to an unrelated DBD.⁴⁸ Overall, however, the MR and glucocorticoid receptors differ markedly in the N-terminal domain, and this region of the receptor is a central determinant of specificity.⁵⁷ Other studies have supported the idea that this domain has functional sequences that limit receptor activity through the recruitment of corepressors, in addition to transcriptional activation functions.^{45,58}

REGULATION OF TRANSCRIPTION INITIATION BY COACTIVATORS AND COREPRESSORS

The major mechanism of MR action is its effects on transcription initiation; however, there may also be effects on transcript elongation.^{59,60} Much has been learned about the generation of an initiation complex and the particular roles that steroid receptors play in this process. Several review articles and book chapters have provided in-depth examinations of the biochemistry of the general transcription machinery, transcription

initiation, promoter escape, and processive elongation.⁶¹⁻⁶³ Most of the coactivators identified so far interact with the C-terminal AF2 domain and include the prototypical GRIP1/TIF2 and SRC,^{64,65} which sequentially recruit a series of different components of the transcriptional machinery and result in the formation of a preinitiation complex. This preinitiation complex includes all the key components of the transcription machinery including RNA polymerase II. A detailed picture of MR-dependent preinitiation complex formation has not been determined. However, the general features are likely similar to those for estrogen receptor⁶⁶ and involve the sequential recruitment by the receptor of the following: 1. chromatin-remodeling SWI/SNF and CARM1/PRTM1 proteins, which promote chromatin remodeling and initiation of complex formation; 2. histone acetylase CBP/P300 (cyclic adenosine monophosphate [cAMP]–responsive element-binding protein), which promotes an active chromatin conformation⁶¹; and 3. direct or indirect recruitment of the TATA-binding protein and other components of the general transcription machinery.⁶⁶

The aforementioned mechanisms are generic and are used by many transcription factors including all steroid receptors, through interactions with the C-terminal AF2 domain. The N-terminal region of the MR, which harbors the AF1 domain, diverges from the other steroid receptors, and other studies have identified coregulators that interact selectively with this receptor domain. ELL (11-19 lysine-rich leukemia factor) is a coactivator for the MR that specifically interacts with AF1b and assists in preinitiation complex formation.⁶⁰ It was originally identified as an elongation factor, and it may also affect transcript elongation. Other specific coregulators include the synergy inhibitory protein PIAS1,⁵⁸ Ubc9,⁶⁷ and p68 RNA helicase.⁶⁸ In many cases, interactions of these regulators with the MR require receptor posttranslational modifications (e.g., by phosphorylation, acetylation, or sumoylation).⁶⁹

REGULATION OF RENAL SODIUM ABSORPTION AND POTASSIUM SECRETION

GENERAL MODEL OF ALDOSTERONE ACTION

Aldosterone effects in the ASDN have been divided into three major phases: latent, early, and late.¹ This designation goes back to the early observations by Ganong and Mulrow that after aldosterone infusion into experimental animals, no effect was observed for at least 15 to 20 minutes.⁷⁰ A similar delay was observed in isolated epithelia.⁷¹ The early phase, which is now known to involve primarily MR-dependent regulation of signaling mediators such as SGK1, culminates in increased apical localization—and, possibly, increased probability of the open state—of the epithelial sodium channel (ENaC). In the late phase, aldosterone stimulates transcription of a variety of effector genes, including those that encode components of the ion transport machinery, notably ENaC and Na⁺-K⁺-adenosine triphosphatase (Na⁺-K⁺-ATPase) subunits. The major direct effect is to increase Na⁺ reabsorption, which is accompanied variably

by Cl^- reabsorption and/or K^+ secretion and, ultimately, water reabsorption. Aldosterone's actions in the principal cells of the connecting segment and collecting duct (Fig. 12.6) are of primary significance; however, this has also been shown to influence fluid and electrolyte transport in other tubule segments, as well as in other organs. These actions of aldosterone can be surmised from the clinical features of individuals with aldosterone-secreting tumors; they have volume expansion with high blood pressure and are commonly (>50% of patients) hypokalemic.^{72,73} In general, the effects of aldosterone on Na^+ absorption

and K^+ secretion work together. However, there are ways whereby these actions can be separated, as discussed later.

$\text{Na}^+\text{-K}^+\text{-ATPase}$, located on the basolateral membrane (blood side), establishes the essential electrochemical gradients that drive ion transport (see Chapters 5 and 6). Importantly, it operates well below its V_{max} , and is seldom, if ever, the rate-limiting step in transepithelial Na^+ transport.⁷⁴ Rather, apical Na^+ entry into the cell via the epithelial Na^+ channel, ENaC, is the rate-limiting step for Na^+ reabsorption by the ASDN and the key locus of regulation.

The discovery of the molecular composition of ENaC in 1993^{75,76} opened the door to understanding how aldosterone

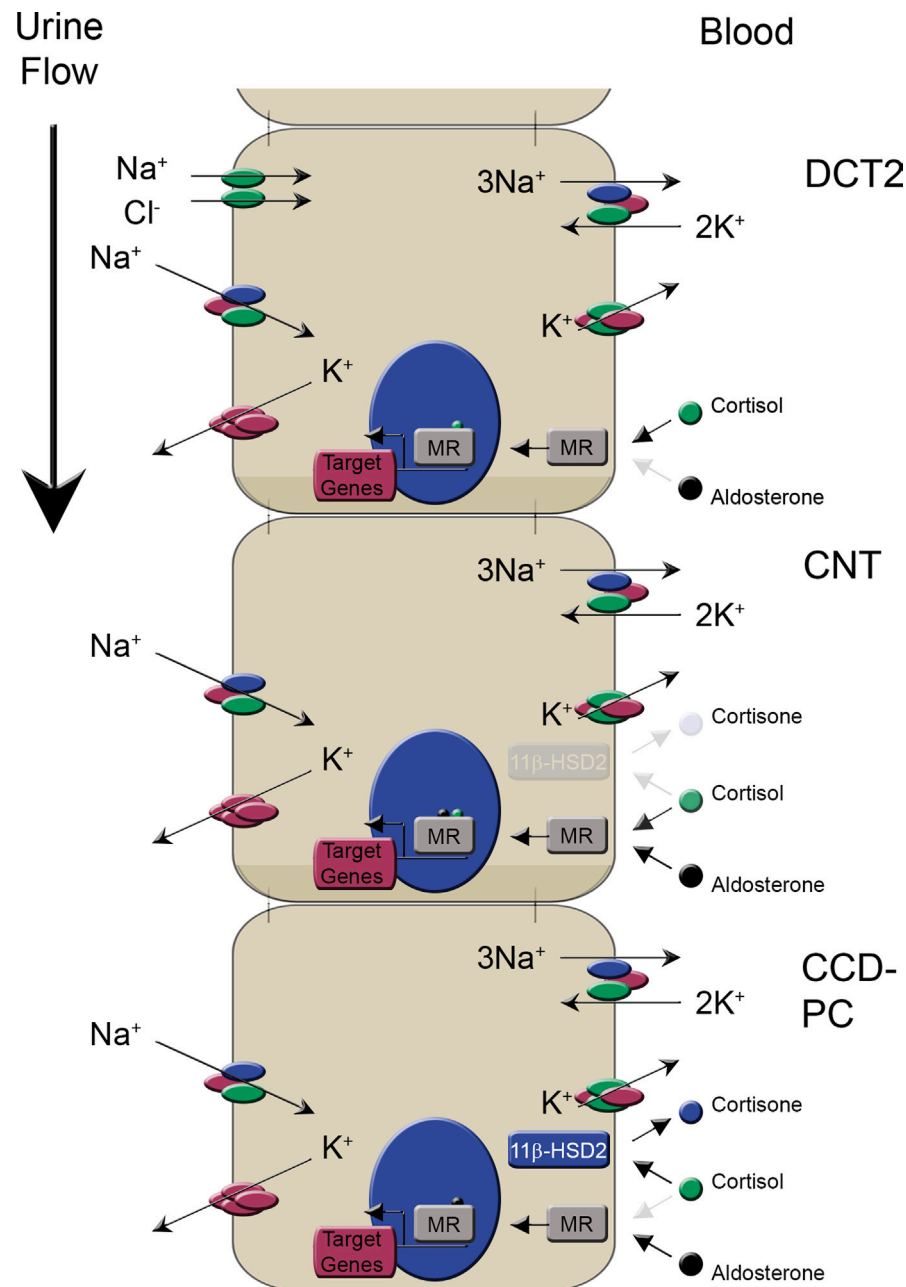


Fig. 12.6 Schematic of principal cells in the aldosterone-sensitive distal nephron (ASDN). The ASDN includes the distal third of the distal convoluted tubule (DCT), connecting tubule, and collecting duct. The $\text{Na}^+\text{-K}^+\text{-adenosine triphosphatase}$ ($\text{Na}^+\text{-K}^+\text{-ATPase}$) establishes the gradients for passive apical entry of sodium through the epithelial sodium channel (ENaC). Transport of sodium through the ENaC creates a negative lumen potential that drives potassium secretion into the lumen. Potassium is also recycled at the basolateral surface, which facilitates potassium exchange across the $\text{Na}^+\text{-K}^+\text{-ATPase}$. Chloride (Cl^-) moves via paracellular and transcellular pathways.

functions to regulate this critically important ion channel. Most Na⁺ transporters are encoded by a single gene product. In contrast, ENaC is composed of three similar but distinct subunits, each encoded by a unique gene. All three subunits come together (probably as a heterotrimer) to form an ion channel with unique biophysical characteristics, the most striking of which is the relatively long time it stays open or closed.⁷⁷ The apical entry of Na⁺ into the cell via ENaC is the rate-limiting step in both Na⁺ absorption and K⁺ secretion.⁷⁸ Na⁺ enters the cell down a steep electrochemical gradient; intracellular [Na⁺] is approximately 10 mmol/L, and the membrane voltage is high (inside negative). Intracellular Na⁺ is pumped out across the basolateral membrane by Na⁺-K⁺-ATPase, as addressed in detail in Chapter 6. Most epithelial cells have a greater density of K⁺ channels on the basolateral membrane and thus recycle K⁺ back into the blood. The distal nephron is unique in that it has an unusually high density of K channels on the apical membrane (primarily Kir 1.1 [renal outer medullary potassium (ROMK)] and BK channels) relative to the apical membrane of other epithelia.^{79,80} This distribution of K⁺ channels permits a large amount of K⁺ that enters the cell via Na⁺-K⁺-ATPase to exit the cell into the lumen and be excreted into the urine. The vast majority of K⁺ that appears in the urine is secreted by the distal nephron.

Much attention has been focused on the early phase of aldosterone action because it appears to be more tractable to dissection, and most changes in Na⁺ current occur during this phase. This separation is probably somewhat artificial, however, and there is considerable overlap in events that define the early and late phases. Moreover, many efforts to manipulate mediators of the early phase (through overexpression and knockdown experiments) have been evaluated after prolonged alteration. Nevertheless, there is some heuristic value in considering the early and late phases of aldosterone action separately.

In cultured collecting duct cells deprived of corticosteroids and then exposed to high concentrations of aldosterone, an increase in ENaC-mediated Na⁺ transport can be observed in well under 1 hour, which is consistent with animal studies.^{70,81} Na⁺ transport continues to increase for 2 to 3 hours, then plateaus for a few hours, and then gradually increases over the next several hours. After 12 hours of exposure to saturating aldosterone concentrations, the increase in ENaC activity is near maximal. The molecular basis for this increase in ENaC activity has been intensively investigated, and several key events are now apparent.

For aldosterone to increase ENaC activity, a change in gene transcription must occur. One of the earliest response genes is *SGK1*.⁸²⁻⁸⁴ This serine-threonine kinase, which mediates a substantial portion of the early effects of aldosterone,^{85,86} is addressed in greater detail later in this chapter, together with its major target, Nedd4-2. The genetic disease Liddle syndrome provided key first clues to ENaC, as addressed further later.^{87,88}

ALDOSTERONE AND EPITHELIAL SODIUM CHANNEL TRAFFICKING

The major action of aldosterone is to increase the number of functional ENaC units on the apical membrane. This process can involve an increase in the number of channel complexes on the surface, and activation of existing complexes, or

both. There is evidence to support both, although the bulk of evidence favors the idea that a change in the number of ENaC units predominates.^{81,89,90} The redistribution of ENaC to the apical membrane can be detected in less than 2 hours after aldosterone exposure.⁸¹

It is less well established whether the number of channels is increased through increased insertion, decreased removal, or both. Aldosterone probably contributes to both processes. Rapid insertion of ENaC is best understood with regard to the actions of cAMP.⁹¹ The extent to which the molecules involved in cAMP-mediated insertion are also involved in aldosterone action is uncertain, but some common mechanisms are probably used. Trafficking to the apical membrane appears to involve hsp70,⁹² SNARE (soluble NEM-sensitive factor attachment protein receptor) proteins,⁹³ and the aldosterone-induced protein melanophilin.⁹⁴ The mitogen-activated protein kinase pathway may also be involved because interruption of extracellular-signal regulated kinase (ERK) phosphorylation by the glucocorticoid-induced leucine zipper (GILZ)⁹⁵ increases ENaC surface expression.

Considerably more is known about how ENaC complexes are retrieved from the apical membrane. This understanding is the direct result of dissecting the molecular consequences of Liddle syndrome, in which mutations in the C-terminus of ENaC lead to increased residence time in the apical membrane.^{87,88} The missing or mutated domains in the β- or γ-subunit of ENaC in this syndrome normally bind to Nedd4-2, a ubiquitin ligase, which ultimately is responsible for initiating endocytosis and degradation.^{96,97} The interaction of Sgk1 and Nedd4-2 in the actions of aldosterone is discussed later. ENaC is internalized via clathrin-coated vesicles, processed into early endosomes, and then further processed into recycling endosomes and late endosomes.^{98,99} Degradation is via lysosomes or proteasomes.^{100,101} The processing of ENaC by vesicular trafficking and its regulation by aldosterone has been reviewed by Butterworth and colleagues.¹⁰²

Phosphatidylinositol-3-kinase (PI3K)-dependent signaling is essential for epithelial Na⁺ transport. It controls SGK1 activity (see later) and also appears to have independent effects on ENaC open probability through direct actions of 3-phosphorylated phosphoinositides, particularly phosphatidylinositol (3,4,5)-trisphosphate.^{103,104} Ras-dependent signaling may also regulate ENaC and the pump in complex ways that depend on downstream signaling through Raf, mitogen-activated protein kinase kinase (MEK), and ERK, as well as through PI3K.¹⁰⁵⁻¹⁰⁹

The late phase of ENaC activation by aldosterone is less well understood than the early phase. A simple evaluation of the late phase is that aldosterone increases the transcription and protein abundance of the ENaC α-subunit. This idea comes from the fact that aldosterone increases the mRNA and protein abundance of α-ENaC in the kidney^{90,110} after a lag of several hours.¹¹¹ Although less well studied, aldosterone appears to produce an increase in β- and γ-subunit expression in the colon.^{111,112} Dietary Na⁺ restriction, a physiologically relevant maneuver that increases aldosterone secretion, clearly increases ENaC surface expression in the renal distal nephron.⁸⁹ However, there appear to be some important differences between chronic aldosterone administration to a Na⁺-replete animal

and chronic dietary Na^+ restriction.^{110,113} Furthermore, it should be noted that increased α -ENaC expression, in and of itself, does not increase ENaC activity in models of collecting duct and lung epithelia, although limiting its expression does restrict aldosterone stimulation.¹¹⁴ It appears that increased expression of α -ENaC may be important for the consolidation of the increase, but it is not sufficient to reproduce the steroid-mediated increase in ENaC activity.

BASOLATERAL MEMBRANE EFFECTS OF ALDOSTERONE

There is general agreement now that whereas the early effects of aldosterone are on apical events, primarily on ENaC, the basolateral and metabolic effects occur later. In addition, the basolateral effects largely result indirectly from the enhanced entry of Na^+ into cells. Notably, increased Na^+ entry has been found to control more than 80% of increased Na^+ - K^+ -ATPase activity and basolateral membrane density in the rat^{115,116} and rabbit¹¹⁵ cortical collecting tubules. Furthermore, striking increases in basolateral membrane folding and surface area occur in aldosterone-treated animals,¹¹⁷ an effect that is markedly attenuated in animals fed a low- Na^+ diet. This result strongly suggests that apical Na^+ entry is required for basolateral changes to occur. However, there is good evidence for direct transcriptional stimulation of Na^+ - K^+ -ATPase subunit expression,^{118,119} as well as reports supporting some direct effects of aldosterone in increasing basolateral pump activity¹²⁰ or at least in constituting the pool of latent pumps, which are then recruited to the basolateral membrane in response to a rise in intracellular $[\text{Na}^+]$.¹²¹

ACTIVATION OF THE EPITHELIAL SODIUM CHANNEL BY PROTEOLYTIC CLEAVAGE

There is now clear evidence that when ENaC is delivered to the apical membrane, both the α - and the γ - but not the β -ENaC subunits can be activated by proteolytic cleavage.

Several proteases can cleave either the α - or γ -ENaC subunits including furin, prostatic, CAP2, kallikrein, elastase, matriptase, plasmin, and trypsin. It is not clear whether activation of ENaC by proteolytic cleavage can be regulated by aldosterone, but the idea certainly has attractive features. If aldosterone could regulate expression of one or more rate-limiting proteases, it would be able to regulate both the number of complexes in the apical membrane and the ability of the channel complex to be active. It appears that aldosterone may regulate the expression of prostatic.¹²² Aldosterone may also regulate the expression of the protease nexin-1 (an inhibitor of prostatic) and other proteases.¹²³

The discovery of ENaC activation by cleavage helps explain how aldosterone might increase ENaC activity by increasing both surface expression and the activity of a single ENaC complex. By phosphorylating Nedd4-2 via SGK1 and reducing its ability to bind to the PY domains of the ENaC subunits, aldosterone increases ENaC residence time on the apical membrane. This additional time permits proteolytic activation by one or more endogenous proteases.¹²⁴

POTASSIUM SECRETION AND ALDOSTERONE

One of the major effects of aldosterone is to increase K^+ secretion (and thus excretion), as depicted in Fig. 12.6. The

key feature of this process—with respect to the direct effects of aldosterone per se—involves the stimulation of Na^+ absorption via ENaC. The dependence of K^+ secretion on Na^+ absorption is the basis of the action of the so-called K^+ -sparing diuretics, amiloride and triamterene, both of which inhibit ENaC.

Increasing ENaC activity produces two major secondary effects that in turn enhance K^+ secretion. First, the enhanced Na^+ conductance of the apical membrane produces depolarization and hence a more favorable electrical driving force for K^+ efflux into the lumen. The second effect relates to the activity of the Na^+ - K^+ pump on the basolateral membrane. The more Na^+ that enters across the apical membrane, the more that must be extruded by the pump. Because the pump operates well below its V_{max} under baseline conditions, a slight increase in intracellular Na^+ concentration markedly stimulates pump activity and more K^+ enters the cell. In isolated, perfused cortical collecting ducts, the amount of secreted K^+ is highly related to the amount of absorbed Na^+ when the stimulus for Na^+ absorption is mineralocorticoid hormone.¹²⁵

Two types of K^+ channels are found in the apical membrane of the ASDN: small conductance (SK, 30–40 picosiemens [pS]) channels encoded by the *ROMK* gene and large conductance (BK, 100–200 pS) channels found in many other cell types, including the apical membrane of the colon. Most of the K^+ channels on the apical membrane of the principal cells appear to be SK, at least as far as can be assessed by patch-clamp analysis. The activity of either channel is not directly increased by aldosterone.^{74,126}

A feature of K^+ secretion is that although apical K^+ channels are abundant in the proximal portion of the ASDN (connecting tubule and cortical collecting duct), they are strikingly less abundant in the medullary collecting duct.^{127–129} Because apical K^+ channels are not regulated by aldosterone, their absence in the medullary collecting duct might uncouple aldosterone-regulated Na^+ reabsorption from K^+ secretion in this segment.

SEPARATION OF SODIUM ABSORPTION AND POTASSIUM SECRETION BY THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

The preceding sections establish a picture that parsimoniously accounts for the effect of aldosterone to stimulate Na^+ reabsorption and K^+ secretion at the same rate. The simple stimulation of electrogenic Na^+ reabsorption (via ENaC) is sufficient to stimulate K^+ secretion, which fits well for organisms faced with a combined low- Na^+ , high- K^+ diet, which was maintained most of the time through millions of years of land vertebrate evolution. However, organisms do not ingest a fixed amount of Na^+ and K^+ , so an inexorable linkage between Na^+ absorption and K^+ secretion by the ASDN cannot possibly occur all the time. Investigators have proposed several possibilities to explain how these processes can be separated.

ROLE OF DISTAL TUBULE FLUID DELIVERY

A traditional view for differential Na^+ and K^+ handling stems from the differing role that aldosterone plays in potassium secretion depending on the tubular flow (and thus sodium

flow) rate. Studies from adrenalectomized dogs have demonstrated that the primary regulators of potassium excretion are the serum potassium concentration and tubular flow rate. A higher serum potassium concentration yields a higher filtered load of potassium. Hyperkalemia also stimulates natriuresis from upstream segments of the nephron.^{130,131} This latter effect can increase tubular flow rate, which, in turn, diminishes potassium concentration in the lumen and activates flow-stimulated BK channel-mediated potassium secretion in the collecting duct. In the setting of sufficient distal delivery of sodium, potassium loading will not yield a higher steady-state concentration of aldosterone because the two mechanisms previously mentioned are sufficient to normalize the serum potassium level.¹³² However, under conditions of sodium depletion, proximal Na⁺ reabsorption is increased, which further diminishes distal delivery of sodium and hence tubular flow rate.¹⁷ This diminishes flow-mediated potassium secretion, so aldosterone secretion is necessary to normalize potassium balance.

INDEPENDENT REGULATION OF SODIUM AND POTASSIUM TRANSPORTERS

Other possible mechanisms have been suggested, which involve separate regulation of sodium and potassium transport (e.g., ENaC and ROMK) by specific stimuli, depending on the state of Na⁺ and K⁺ intake. With a constant Na⁺ intake, one could envision that a high-potassium diet could enhance the activity of ROMK, whereas a low-potassium diet would reduce its activity. Such an effect would cause more or less of the K⁺ entering the cell via the Na⁺-K⁺ pump to be recycled across the basolateral membrane. This mechanism, although probably complex in its execution, is appealing in its simplicity.

ROLE OF WNKs

The regulatory roles of the WNK (With-No-Lysine [K]) family of kinases, which are discussed in detail in [Chapter 6](#), are complex and remain controversial. It is clear that wild-type WNK4 stimulates the Na⁺,Cl⁻ cotransporter (NCC) and that a K⁺-induced increase in intracellular [Cl⁻] plays a key role in normal regulation. However, the roles of WNK1 and WNK4 in regulating ion channels in the ASDN such as ROMK and ENaC are more controversial.^{133,134} It is currently thought that in the DCT, low plasma (and hence extracellular) K⁺ concentration hyperpolarizes the basolateral membrane via Kir4.1/5.1, which leads to a reduction in intracellular chloride, thereby activating WNK4 and SPAK (STE20/SPS1-related proline-alanine-rich kinase) and hence NCC phosphorylation.^{135,136} A high plasma K⁺ concentration has the opposite effect and inhibits NCC phosphorylation; however, the mechanism remains more controversial and may be chloride-independent. Thus through its effects to raise the plasma K⁺ concentration, a high-K⁺ diet has effects both locally in the kidneys and through aldosterone to stimulate K⁺ excretion. Aldosterone acts directly in the ASDN to stimulate ENaC (in part through SGK1), which increases the driving force for K⁺ secretion through ROMK and BK channels.¹³⁷ In conjunction, an elevated K⁺ concentration acts directly in DCT cells to inhibit NCC phosphorylation

and enhance delivery of Na⁺ to the ENaC-expressing segments. According to this view, aldosterone does not have a direct effect on NCC-mediated electroneutral Na⁺ transport.

ROLE OF CHLORIDE TRANSPORT REGULATION

There is also evidence supporting the independent regulation of Cl⁻ transport in the collecting duct. Cl⁻ can be absorbed by the paracellular pathway (i.e., between cells) driven by the lumen-negative voltage across the epithelium. This pathway can be influenced by aldosterone.¹³⁸ Cl⁻ can also be absorbed through the cells by specific transporters. One example of a Cl⁻ transporter in the collecting duct is pendrin, an anion exchanger present on the apical membrane of intercalated cells. Mice that lack this transporter do not tolerate NaCl restriction as well as normal mice.¹³⁹ Its activity is dependent on Cl⁻ delivery to the distal nephron, and it is coregulated by angiotensin II and aldosterone.¹⁴⁰⁻¹⁴²

As discussed in the next section, in contrast to principal cells, the ability of the MR in intercalated cells to respond to aldosterone is regulated by the phosphorylation of Ser-843 in its ligand-binding domain. It should also be noted that modulation of Na⁺ absorption in the medullary collecting duct may also play a role in the balance of Na⁺ reabsorption and K⁺ secretion; this segment has little capacity to secrete K⁺, and endogenous paracrine factors such as prostaglandins E₂ and transforming growth factor- β , which have potent inhibitory effects on Na⁺ transport, are increased in response to a high-NaCl diet.^{143,144}

DIFFERENTIAL REGULATION OF INTERCALATED CELL MINERALOCORTICOID RECEPTOR

It has become increasingly clear that intercalated cell Cl⁻ transport contributes to collecting duct NaCl absorption, and this plays an important role in allowing distinct responses to aldosterone in states of volume depletion versus hyperkalemia.^{10,11} A central feature of this proposed regulation is differential phosphorylation of the MR LBD in intercalated cells, as shown schematically in [Fig. 12.7](#). When phosphorylated at S843 in the LBD, the MR cannot bind aldosterone (or cortisol) and thus cannot be activated. This phosphorylation, which is stimulated by hyperkalemia, occurs selectively in intercalated cells but not in principal cells. Angiotensin II, on the other hand, induces S843 dephosphorylation in intercalated cells, markedly increasing ligand binding and therefore activation. Intercalated cells are known predominantly to mediate H⁺ transport; however, other studies have implicated them in electroneutral NaCl transport via the combined actions of the Na⁺-dependent Cl⁻-HCO₃⁻ exchanger (NDCBE)¹⁴⁵ and the apical Cl⁻-HCO₃⁻ exchanger, pendrin.^{146,147} Thus when the MR is active in these cells (S843 dephosphorylated), electroneutral NaCl transport occurs, without enhancing the driving force for K⁺ secretion. Because intercalated cells lack 11 β -HSD2 under these conditions, it is cortisol that binds to and activates the MR. When the intercalated cell MR is inactive (S843 phosphorylated), aldosterone acts in principal cells to stimulate ENaC-dependent electrogenic Na⁺ transport, which enhances K⁺ secretion.

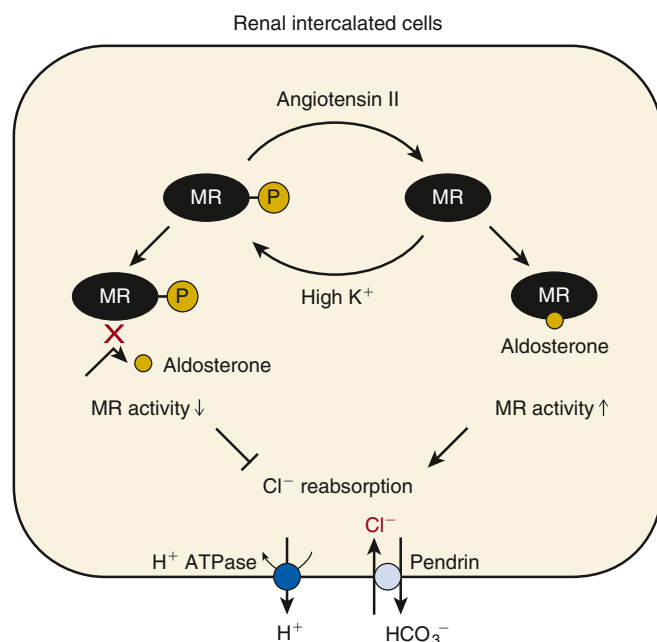


Fig. 12.7 The role of mineralocorticoid receptor (MR) ligand-hormone-binding domain (LBD) phosphorylation in controlling chloride reabsorption by intercalated cells. When phosphorylated at Ser-843 in the LBD, MR cannot bind ligands and hence cannot be activated. This phosphorylated state of MR is found only in intercalated cells, not in neighboring principal cells. In states of volume depletion, an elevated angiotensin II level decreases MR phosphorylation at Ser-843 (right side), allowing activation. In intercalated cells, MR mediates stimulation of both the proton pump (H^+ -ATPase) and Cl^- - HCO_3^- exchangers such as pendrin, thereby increasing Cl^- reabsorption and promoting increased plasma volume while inhibiting K^+ secretion. In contrast, in states of hyperkalemia, phosphorylation of Ser-843 is increased (left side) and hence Cl^- reabsorption by intercalated cells is decreased and the principal cell-dependent K^+ secretion is increased. (From Shibata S, Rinehart J, Zhang J, et al. Regulated mineralocorticoid receptor phosphorylation controls ligand binding and renal response to volume depletion and hyperkalemia. *Cell Metab.* 2013;18:660–671.)

ALDOSTERONE-INDEPENDENT ENaC-MEDIATED SODIUM REABSORPTION IN THE DISTAL NEPHRON

The term “aldosterone-sensitive distal nephron” emphasizes the central role of this steroid in the control of ion transport in this region of the nephron. However, ENaC activity and aldosterone sensitivity exhibit axial heterogeneity from the late distal convoluted tubule (DCT2) through the connecting tubule to the cortical collecting duct and, finally, to the medullary collecting duct. In mice on a standard sodium diet, total ENaC expression increases with progression from the DCT2 to the connecting tubule,¹⁴⁸ although ENaC apical localization and activity are higher in the DCT2. Only under conditions of a low-sodium diet or aldosterone administration does the importance of the connecting tubule in particular, and to a lesser extent the cortical collecting duct, emerge. The total luminal surface area in the DCT2 and connecting tubule is several-fold higher than in the cortical collecting duct,¹⁴⁹ and together these two segments appear to be sufficient to maintain sodium balance, even in the absence of detectable

ENaC along the collecting duct. Mice lacking ENaC selectively in the collecting duct come into balance, even on a low-sodium diet.¹⁵⁰ Congruent with these findings, deletion of α -ENaC from the DCT2, connecting tubule, and collecting duct results in severe sodium wasting.¹⁵¹ Notably, it is the connecting tubule that appears to be most important in the response to aldosterone, whereas DCT2 has the highest baseline transport in the absence of aldosterone.¹⁵² The cortical collecting duct is not as critical as was originally thought for either baseline or aldosterone-stimulated sodium reabsorption, probably due to its smaller surface area. Data have established a key role of aldosterone-independent (likely cortisol-dependent) activation of MR in regulating ENaC in this context.^{153,154} Importantly, although the later part of the DCT (DCT2) and early CNT express MR, ENaC, and ROMK, they express relatively low levels of 11 β -HSD2, and hence MR is activated by cortisol to stimulate K^+ secretion. Evidence has also begun to establish a mechanism for aldosterone-independent effects of K^+ within principal cells: Local changes in $[\text{K}^+]$ within the kidney stimulate SGK1 phosphorylation by mTORC2,^{155,156} possibly through a WNK-dependent mechanism, as addressed in Chapter 6).¹⁵⁷ Rapid activation of ENaC then drives ROMK-mediated K^+ secretion.

As we continue to traverse the nephron, further sodium reabsorption is minimal in the medullary collecting duct, on a normal sodium diet, and is not significantly stimulated by aldosterone.¹²⁵

SITES OF MINERALOCORTICOID RECEPTOR EXPRESSION AND LOCUS OF ACTION ALONG THE NEPHRON

ALDOSTERONE-SENSITIVE DISTAL NEPHRON

In the kidney, the MR is expressed at the highest levels in distal nephron cells extending from the last third of the DCT through the medullary collecting duct,¹⁵⁸ which is frequently referred to as the ASDN (Fig. 12.8).¹ This pattern of expression was first demonstrated using labeled hormone-binding studies performed before the cloning of the MR¹⁵⁹ and has been confirmed since by several methods including the polymerase chain reaction assay,¹⁶⁰ in situ hybridization,¹⁶¹ and immunohistochemical analysis.¹⁶² Effects of aldosterone on electrogenic Na^+ and K^+ transport in principal cells have been found consistently in these nephron segments,¹⁵⁸ which also express ENaC, and 11 β -HSD2, as addressed in detail earlier.

Collecting duct-intercalated cells also express the MR and respond specifically to aldosterone and alter proton secretion. Aldosterone directly increases the activity of the H^+ -ATPase in the collecting duct, and its absence results in decreased proton secretion.^{163–165} Interestingly, nongenomic stimulation of H^+ -ATPase activity in type A intercalated cells has been demonstrated in isolated murine collecting ducts.¹⁶⁶ Consistent with these effects, aldosterone deficiency results in distal renal tubular acidosis type 4, and excess aldosterone results in metabolic alkalosis.¹⁶⁷ It should be noted that aldosterone also stimulates H^+ secretion due to effects on principal cell Na^+ transport, which alters the electrical gradient. These older studies must also now be interpreted in the context of more recent data,¹⁰ which, as noted earlier,

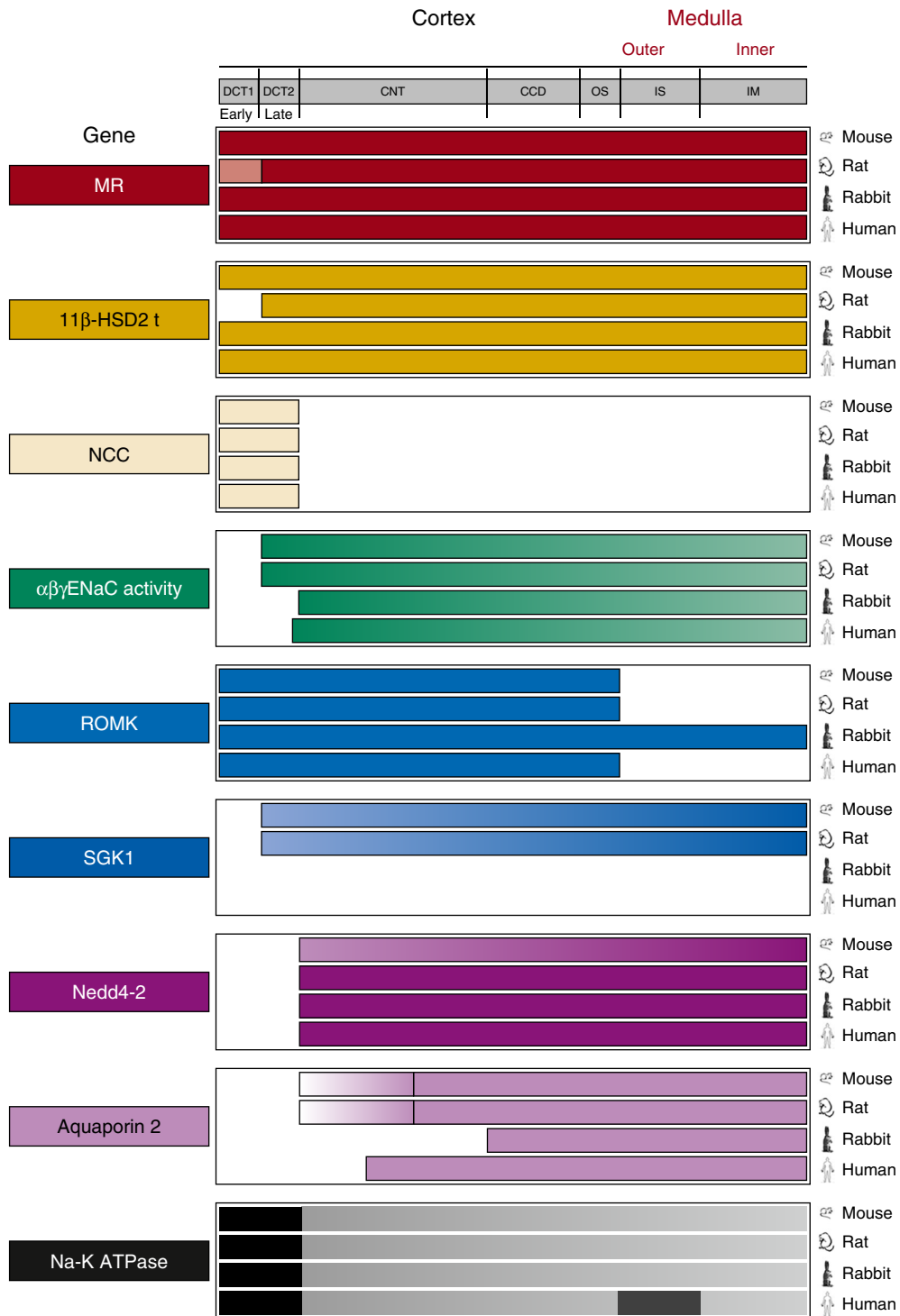


Fig. 12.8 Expression and/or activity of the mineralocorticoid-dependent transport machinery in principal cells along the mature aldosterone-sensitive distal nephron (ASDN). Mineralocorticoid specificity is conferred by the presence of the mineralocorticoid receptor (*MR*) and 11β-hydroxysteroid dehydrogenase type 2 (*11β-HSD2*), beginning primarily from the latter part of the distal convoluted tubule (*DCT*). The thiazide-sensitive sodium chloride cotransporter (*NCC*) is expressed exclusively in the DCT, but after the transition from the DCT to the connecting tubule (CNT), sodium reabsorption is distinctly determined by amiloride-sensitive sodium channel (*ENaC*) activity. *ENaC* activity is strongest in the CNT and decreases down to the inner medulla collecting duct. Variation in gene expression or activity along the nephron is indicated by the intensity of shading. Note that there is some variation in gene expression from mouse to human. However, the machinery for sodium reabsorption in the ASDN is predominantly conserved across species. Each nephron segment is drawn to scale, but expression of channels and transporters in intercalated cells is omitted. Expression and/or activity is based on messenger RNA, protein, and biochemical studies. G, Glomerulus; *PCT*, proximal convoluted tubule; *ROMK channel*, renal outer medullary potassium; *SGK1*, serum- and glucocorticoid-regulated kinase 1. (Modified from Loffing J, Korbmayer C. Regulated sodium transport in the renal connecting tubule [CNT] via the epithelial sodium channel [ENaC]. *Pflügers Archiv*. 2009;458[1]:111–135.)

demonstrate that the effect of aldosterone on intercalated cells depends on the genesis of the signal.

OTHER SITES OF EXPRESSION

The MR has been identified at some level in all parts of the nephron examined, including the glomerulus.^{160,168-173} Its effects, at least at some of these sites, are likely to be physiologically relevant in states of volume depletion and acid-base disturbances; however, the data are not as robust and consistent as those for the ASDN.

Glomerulus

The MR (but not 11 β -HSD2) is expressed in glomerular mesangial cells, where it is thought to affect proliferation and production of reactive oxygen species^{174,175} and to have profibrotic effects through SGK1.¹⁷⁶ These effects have been suggested to be important in the progression of renal damage, particularly in diabetic nephropathy,¹⁷⁷ in which glucocorticoids mimic the activity of aldosteronism in the context of tissue damage. However, the physiologic role of mesangial cell MR is uncertain.

Proximal Convoluted Tubule

Hierholzer and Stolte have shown, through elegant microperfusion studies, that the sodium reabsorptive capacity of the proximal convolution is decreased in adrenalectomized animals and restored by administration of aldosterone.¹⁷⁸ Chronic volume depletion increases sodium reabsorption in the proximal convoluted tubule, which is in part mediated by the MR. The mechanisms of action in this nephron segment are controversial. Some studies have indicated a MR-dependent increase in the activity of Na⁺-H⁺-exchanger isoform 3, possibly through an increase in trafficking of the transporter to the membrane.¹⁷⁹⁻¹⁸² This transporter contributes to sodium and bicarbonate reabsorption. MR activation, in turn, may activate the Na⁺-K⁺-ATPase in the basolateral membrane of the proximal convoluted tubule to maintain a gradient for sodium reabsorption.¹⁸³⁻¹⁸⁶

Medullary Thick Ascending Limb

In the medullary thick ascending limb, mineralocorticoids but not glucocorticoids increase sodium and chloride reabsorption. In rodents, adrenalectomy impairs the reabsorption of NaCl in the medullary thick ascending limb, and aldosterone restores this process.^{187,188} This reabsorptive defect may contribute to the urinary concentrating and diluting abnormality measured in patients with Addison disease and in mice lacking aldosterone synthase.^{9,178,189} The medullary thick ascending limb also participates in the regulation of acid-base balance by reabsorbing most of the filtered HCO₃⁻ that is not reabsorbed by the proximal tubule. In this context, aldosterone has been shown to stimulate the Na⁺-H⁺ exchanger in the amphibian thick ascending limb, possibly through a rapid nongenomic effect.¹⁹⁰ Other studies have also implicated regulation of the Na-K-2Cl cotransporter type 2 in the thick ascending limb—as well as the NCC in the DCT (see later)—by oxidative stress response kinase 1 (OSR1) and STE20/SPS1-related proline-alanine-rich kinase (SPAK) (OSR1/SPAK).^{191,192}

Distal Convoluted Tubule

The studies of potassium regulation described earlier have also provided insight into the role of direct or indirect aldosterone-induced NCC transport. Aldosterone increases NCC phosphorylation and total protein¹⁹³ abundance but, until recently, the mechanism of this upregulation was unclear. Several groups have shown that aldosterone indirectly stimulates NCC via the activation of ENaC-mediated sodium transport and the resultant potassium secretion^{194,195} and hypokalemia. In turn, a lower plasma potassium level activates NCC.^{135,136} Another potential mechanism is through direct regulation by aldosterone. In rodent studies, aldosterone stimulates serum and glucocorticoid kinase 1, which inhibits the ubiquitin ligase Nedd4-2, which, in turn, can regulate WNK1 and NCC phosphorylation.¹⁹⁶ As discussed later, this pathway mirrors a well-known mechanism of aldosterone-mediated disinhibition of Nedd4-2 and subsequent ENaC degradation in principal cells. The specific physiologic contexts for these different direct and indirect modes of aldosterone-dependent NCC are still unknown.

NONRENAL ALDOSTERONE-RESPONSIVE TIGHT EPITHELIA

The mineralocorticoid effects of aldosterone have predominantly been studied in the distal nephron but do influence other—mostly ENaC-expressing—tight epithelia. ENaC is present in visceral epithelial cells of the distal colon, distal lung, salivary glands, sweat glands, and taste buds.

COLON

Under physiologic conditions, approximately 1.3 to 1.8 L of electrolyte-rich fluid is reabsorbed per day from the colonic epithelium.¹⁹⁷ Like the nephron, the proximal colon reabsorbs sodium via an electroneutral, ENaC-independent process. In the distal colon, electrogenic Na⁺ absorption via ENaC channels is the predominant mode of sodium transport.¹⁹⁸⁻²⁰¹ In the colon, as in the distal nephron, MR signaling is aldosterone selective, reflecting the activity of 11 β -HSD2.²⁰² Aldosterone increases electrogenic sodium absorption and potassium secretion and inhibits electroneutral absorption.²⁰³ This is in contrast to glucocorticoids, which, at higher concentrations, activate glucocorticoid receptors to stimulate electroneutral absorption in the proximal and distal colon.^{78,204} As in the distal nephron, the aldosterone response can be characterized by an early and late response. The early response gene, *SGK1*, is upregulated by aldosterone via MR.⁸⁴ However, in contrast with the kidney, aldosterone and a low-salt diet have been shown to stimulate transcription of β -ENaC but not α -ENaC in rat models.^{205,206}

Aldosterone stimulates electrogenic potassium secretion from colonic epithelia. The significance of this secretion is evident in anuric patients. Potassium secretion from the colon is much higher in patients undergoing long-term hemodialysis than in patients not undergoing dialysis.²⁰⁷⁻²⁰⁹ Indeed, administration of fludrocortisone, a mineralocorticoid agonist, to dialysis patients has been

shown to reduce hyperkalemia in small clinical trials.²¹⁰ Low doses of the common MR antagonist spironolactone do not result in significant hyperkalemia.²¹¹⁻²¹³

LUNG

Vectorial transport of salt and water across the distal airway epithelium and alveoli primarily determines fluid clearance from the lung. ENaC is the rate-limiting step in sodium transport in the lung and plays a primary role in several physiologic and pathophysiologic conditions determined by fluid clearance.²¹⁴ The molecular apparatus for mineralocorticoid-stimulated fluid reabsorption via ENaC (concomitant MR and 11 β -HSD2) is present in late gestational and mature adult lung in humans^{202,215,216} and rats,²¹⁷ and there is some evidence for a significant physiologic role of aldosterone in ENaC-mediated sodium transport,²¹⁷ although glucocorticoids acting via the glucocorticoid receptor are likely to play the predominant role in lung.^{84,110,218-221} Importantly, glucocorticoids, but not mineralocorticoids, play a critical role in lung maturation in humans, and glucocorticoid receptor knockout mice, like α -ENaC knockout mice, die of respiratory insufficiency within hours of birth. In contrast, MR knockout mice demonstrate a severe salt-wasting phenotype but no significant lung phenotype.^{163,222}

EXOCRINE GLANDS AND SENSORS

ENaC-mediated sodium reabsorption is also measurable in the salivary and sweat glands.²²³ The importance of these tissues for sodium and water homeostasis is underscored by rare genetic mutations that result in elevated plasma aldosterone levels and pseudohypoaldosteronism, with normal renal tubular function but significant sodium loss from salivary or sweat glands.^{224,225} ENaC channels also play an important role in transduction of sodium salt taste in the anterior papillae of the tongue.^{226,227} The appropriate molecular machinery for mineralocorticoid-responsive sodium reabsorption is expressed in these organs.^{202,228,229} As in colonic epithelia, aldosterone stimulates the expression of β - and γ -ENaC and sodium transport in glands and taste buds in animal models.^{226,230} Moreover, in humans, changes in dietary sodium are inversely proportional to sodium transport across salivary epithelia.²³¹ Similar to the aldosterone-responsive distal nephron and distal colon, sodium uptake is coupled with potassium secretion in salivary epithelia. This effect is evident in humans with hyperaldosteronism. Such patients have a salivary [Na⁺]/[K⁺] ratio significantly lower than that of subjects without the disorder,^{232,233} although this has not been accepted as a valid means to screen for hyperaldosteronism.

ROLE OF SERUM- AND GLUCOCORTICOID-REGULATED KINASE IN MEDIATING ALDOSTERONE EFFECTS

INDUCTION OF SGK1 BY ALDOSTERONE

The serine-threonine kinase, SGK1, is the best characterized transcriptional target of MR.^{83,84} Its mRNA levels are increased within 15 minutes, and protein levels within 30

minutes by aldosterone in both cultured cells and in native collecting duct.^{81,84,186,185,234} (notably, more strongly in CNT and cortical collecting duct than in the medulla^{152,235}). SGK1 is expressed in other nephron segments including glomeruli, proximal tubule, and papillae^{81,84,236}; however, its rapid induction in the ASDN appears to provide most of the basis for its role in aldosterone-regulated sodium and potassium transport. SGK1, induced by high sodium in infiltrating T cells, is also important for inflammation in the kidney and hypertension, although the specific nephron segments are unknown.^{180,467} Although it has effects on proliferation and apoptosis in kidney cells, these effects appear to be minor and the control of ENaC and other transporters²³⁷ predominates.

MOLECULAR MECHANISMS OF SGK1 ACTION IN THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

SGK1 gene transcription is induced by a variety of stimuli in addition to MR. As its name implies, these include serum and glucocorticoids but also follicle-stimulating hormone, transforming growth factor- β , and osmotic stress.²³⁸⁻²⁴¹ Some of these, particularly osmotic regulation, may play significant roles in renal physiology. Stimulation of SGK1 kinase activity is primarily through phosphorylation,²⁴²⁻²⁴⁶ which, like that of its close relative, Akt, is stimulated by a variety of growth factors including insulin and insulin-like growth factor-1²⁴⁵⁻²⁴⁷; these act through PI3K to trigger phosphorylation at two key residues, an activation loop (residue T256) and a hydrophobic motif (S422). PI3K-dependent kinase 1 (PDK1) phosphorylates T256, and mammalian target of rapamycin [mTOR] complex 2 (mTORC2) phosphorylates S422.^{242,245,248-251} Thus SGK1 serves as a convergence point for different classes of stimuli, which act on the one hand to control its expression (aldosterone and cortisol) and on the other to control its activity (insulin and other activators including K⁺), which results in the coordinate regulation of ENaC.

In the study of the physiologic and pathophysiologic roles of SGK1 in the ASDN, mice lacking SGK1 under different physiologic stimuli have provided considerable insight. Unlike MR knockout mice,¹⁶³ mice lacking SGK1 survive the neonatal period and appear normal when consuming a normal sodium/normal potassium diet, albeit with markedly elevated aldosterone. When subjected to a low-sodium diet, these mice have marked sodium wasting, and in the face of a high-potassium diet they develop hyperkalemia, akin to pseudohypoaldosteronism type I.²⁵²⁻²⁵⁴ Additional mouse models of SGK1 deletion have demonstrated diminished processing of ENaC subunits I.^{253,255} Notably, this is a significantly milder phenotype than with deletion of MR or α -ENaC.²⁵⁶ These comparisons suggest that disruption of SGK1 signaling may be insufficient to eliminate aldosterone-mediated sodium transport due to additional aldosterone-induced and aldosterone-repressed proteins, which could compensate for the lack of SGK1.

SGK1 may also play a significant role in states of aldosterone excess or upregulation of hormonal activators of SGK1 (e.g., insulin). SGK1 knockout mice are protected from the development of salt-sensitive hypertension, which accompanies the hyperinsulinemia of the metabolic

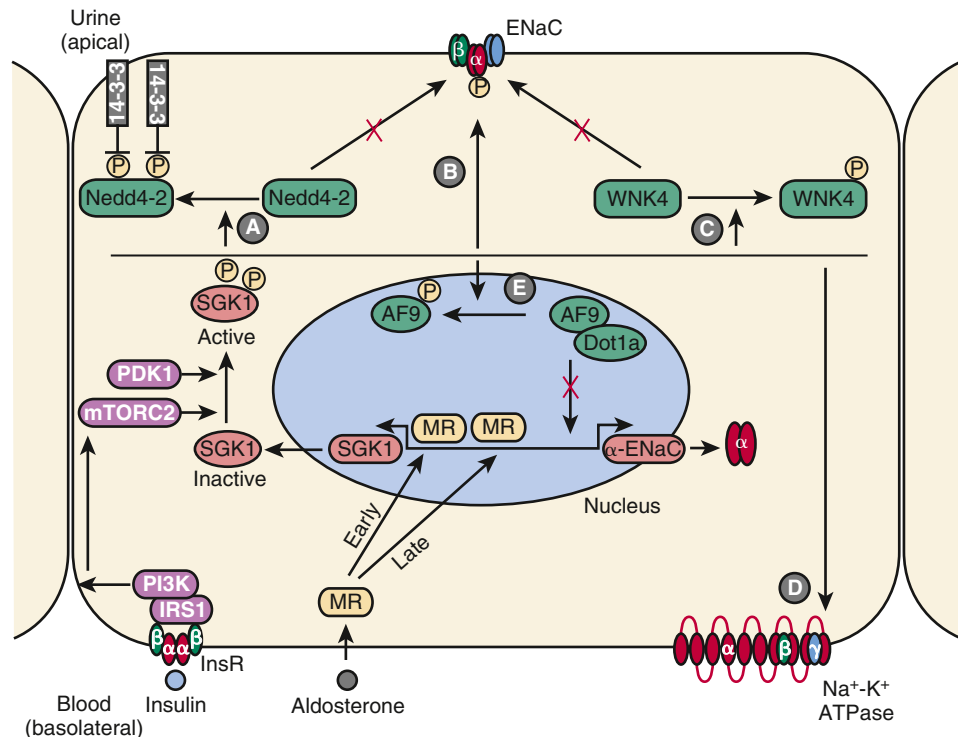


Fig. 12.9 Mechanisms of serum- and glucocorticoid-regulated kinase 1 (SGK1)-mediated stimulation of the amiloride-sensitive sodium channel (ENaC). Within principal cells of the mammalian kidney, SGK1 is transcriptionally upregulated as an early aldosterone-induced gene product. SGK1 is then phosphorylated twice via a phosphatidylinositol-3-kinase (PI3K)-dependent cascade of upstream kinases leading to active SGK1. Active SGK1 has multiple effects: It increases apical plasma membrane ENaC by inhibiting Nedd4-2 and Raf-1, and it induces transcription of the α -ENaC (thereby influencing late effects of aldosterone). (A–E, clockwise) Shown are the individual mechanisms that have been elucidated in principal cells. See text for details. *InsR*, Insulin receptor; *IRS1*, insulin receptor substrate 1; *MR*, mineralocorticoid receptor; *mTORC2*, mammalian target of rapamycin complex 2; *PDK1*, 3-phosphoinositide-dependent protein kinase type 1.

syndrome.^{257,258} Taken together, SGK1 is an important component of ENaC regulation to maintain both sodium and potassium homeostasis.

Despite its accepted role as a mediator of aldosterone-stimulated sodium reabsorption, the mechanisms whereby SGK1 stimulates ENaC are not fully characterized. Several mechanistic studies have demonstrated that SGK1 is rapidly induced but also rapidly degraded.^{84,259,260} The N-terminus of the kinase, which distinguishes SGK1 from other kinase family members (e.g., Akt), is important for stimulation of sodium transport but is also the target for rapid degradation of the kinase via the ubiquitin-proteasome system.^{261–267} The molecular mechanisms of ENaC stimulation by SGK1 can be divided into three known categories (Fig. 12.9): 1. posttranslational effects on the E3 ubiquitin ligase Nedd4-2; 2. posttranslational Nedd4-2-independent effects; and 3. transcription of gene products such as α -ENaC.

SGK1 INHIBITS THE UBIQUITIN LIGASE Nedd4-2

Neural developmentally downregulated isoform 4-2 (Nedd4-2) is an E3 ubiquitin ligase, which interacts particularly with the C-terminal tails of α -ENaC, β -ENaC, and γ -ENaC²⁶⁸ and decreases channel surface expression via ubiquitination and internalization.^{101,269} The genetic defects in Liddle syndrome (ENaC-mediated hypertension, hypokalemia, and metabolic alkalosis) are due to mutations in the C-terminal tail of ENaC subunits (primarily α and γ), which result in decreased inhibition by Nedd4-2 and hence increased ENaC activity.²⁷⁰ Lack of Nedd4-2 in vivo results in

increased ENaC activity and salt-sensitive hypertension,^{271,272} recapitulating a Liddle syndrome-like phenotype. SGK1 interacts with and phosphorylates Nedd4-2^{243,273} in an ENaC signaling complex¹⁰⁷ and enhances cell surface expression of ENaC^{274,275} (see Fig. 12.9A). This interaction coordinates the phosphorylation-dependent binding of 14-3-3 proteins to inhibit Nedd4-2 and prevent the ubiquitination of ENaC.^{276–279} This disinhibition of ENaC reflects a recurring theme in the regulation of ion transport in the kidney seen with the WNK kinases and NCC, other aldosterone-regulated gene products (e.g., GILZ) and ENaC, and sodium-hydrogen exchange regulatory cofactor 2 (NHERF2) and ROMK.^{280,281}

SGK1 ENHANCES EPITHELIAL SODIUM CHANNEL ACTIVITY INDEPENDENTLY OF Nedd4-2

In cell culture systems, mutation of SGK1 phosphorylation sites on Nedd4-2 does not completely abolish the ability of SGK1 to stimulate ENaC.²⁴³ Furthermore, SGK1 has been shown to stimulate ENaC channels with Liddle syndrome mutations, which are unable to bind Nedd4-2.^{84,274} Consequently, other Nedd4-2-independent mechanisms of SGK1 stimulation have been proposed. SGK1 directly phosphorylates a serine residue in the intracellular C-terminal tail of α -ENaC, which directly activates channels at the cell surface (see Fig. 12.9B).^{282,283} SGK1 has been implicated in the stimulation of ENaC via the phosphorylation of WNK4, a kinase mutated in familial hyperkalemic hypertension (FHt)/pseudohypoaldosteronism type II (PHA II) (see Fig. 12.9C).²⁸⁴ Cell surface-expressed SGK1

may also increase open probability of the channel.^{283,285} In addition to showing effects on ENaC, SGK1 has been found to stimulate the activity of basolateral Na⁺-K⁺-ATPase, which separately increases ENaC-mediated sodium transport (see Fig. 12.9D).^{286,287} The time course of these effects and their relative importance compared with Nedd4-2-dependent inhibition have not been explored.²⁸² The next generation of molecular studies of SGK1 will elucidate the relative importance of each of these pathways.

SGK1 STIMULATES EXPRESSION OF COMPONENTS OF SODIUM TRANSPORT MACHINERY

SGK1 also regulates the expression of late aldosterone-responsive genes, primarily α -ENaC.^{288,289} Active SGK1 is an important mediator of aldosterone-sensitive α -ENaC transcription *in vivo* via inhibition of a transcriptional repression element, the disruptor of telomeric silencing alternative splice variant a (Dot1a)-ALL1-fused gene from the chromosome 9 (Af9) complex.²⁸⁹ SGK1 phosphorylates Af9 and reduces the interaction between Dot1a and Af9. This releases suppression of ENaC transcription by this complex (see Fig. 12.9E). Thus SGK1 acts on ENaC channels to enhance sodium channel activity rapidly through the increase of active channels at the apical surface and the increase of Na⁺-K⁺-ATPase at the basolateral surface and also stimulates the transcription of elements of the machinery for sodium transport to promote a sustained response to aldosterone. SGK1 is an early-onset gene, but its effects influence both immediate- and long-term aldosterone-stimulated sodium reabsorption.

SGK1 STIMULATES POTASSIUM SECRETION IN THE ALDOSTERONE-SENSITIVE DISTAL NEPHRON

Further evidence of a role for SGK1 in the regulation of sodium transport in the ASDN has been revealed by the study of potassium secretion. If SGK1 enhances ENaC-mediated sodium transport, the potential difference across the apical to basolateral surface of principal cells should be higher (more negative) and thus should indirectly stimulate potassium secretion. SGK1 knockout mice are unable to secrete potassium adequately when challenged with a high-potassium diet, and mice with constitutive or inducible deletion of SGK1 are prone to hyperkalemia.^{255,290,291} Moreover, the potential difference across collecting duct epithelia from these knockout mice indicates that the effect of SGK1 on potassium secretion occurs via ENaC, not through direct regulation of ROMK.²⁹⁰ SGK1 also directly inhibits Nedd4-2, and deletion of Nedd4-2 predisposes low-potassium-fed mice to hypokalemia via the constitutive stimulation of ENaC-mediated sodium transport.²⁹² Thus SGK1 and its effectors have physiologically relevant roles in sodium and potassium transport in the ASDN.

ALTERNATE MODES OF REGULATION OF ENaC-MEDIATED SODIUM TRANSPORT BY ALDOSTERONE

Early aldosterone-induced mRNAs other than SGK1 have also been implicated in the stimulation of ENaC; these include K-ras, GILZ, kidney-specific WNK1, Usp45, melanophilin, and promyelocytic leukemia zinc finger.^{94,95,293-296}

Evidence has also supported a key role of aldosterone-regulated micro-RNAs in regulating both SGK1 and

ENaC.²⁹⁷ Aldosterone can upregulate or downregulate several micro-RNAs in cultured cells and *in vivo*. These micro-RNAs can then indirectly increase or decrease protein levels of intermediate regulators of ENaC-mediated transport. The first micro-RNA cluster (mmu-miR-335-3p, mmu-miR-290-5p, and mmu-miR-1983) to be described by Butterworth and colleagues^{298,299} was downregulated by aldosterone within 24 hours and thereby released the 3' untranslated region of ankyrin 3 to increase apical trafficking of α -ENaC. Aldosterone also increases micro-RNAs that inhibit a negative regulator of ENaC, intersectin 2.²⁹⁷ Aldosterone has also been shown to promote the rapid induction of SGK1 mRNA by decreasing micro-RNA 466g in cultured cells.³⁰⁰ Taken together, SGK1 plays a prominent role in transducing the effect of aldosterone to stimulate ENaC for both regulation of blood pressure and potassium homeostasis. Additionally, there are alternate effectors of aldosterone, but the physiologic contexts of these other pathways have not been as well established.

11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2

ESSENTIAL DETERMINANTS OF MINERALOCORTICOID SPECIFICITY

The physiologic glucocorticoid, cortisol (corticosterone in rats and mice), has a high affinity for the MR, equivalent to that of aldosterone, and, as noted earlier, circulates at plasma-free concentrations that are 1000-fold or more higher than those of aldosterone. Central to the ability of the MR to respond to aldosterone selectively in the ASDN is coexpression of the enzyme 11 β -HSD2.^{4,5} 11 β -HSD2 converts cortisol (corticosterone) to receptor-inactive 11-keto steroids (cortisone in humans, 11-dehydrocorticosterone in rats and mice), using nicotinamide adenine dinucleotide (NAD) as a cosubstrate and generating sufficient amounts of the reduced form of NAD (NADH) to alter the redox potential of the cell. This dependence sets it in contrast to 11 β -HSD1, which uses the reduced form of nicotinamide adenine dinucleotide phosphate (NADPH), preferentially catalyzes the conversion of glucocorticoids from the oxidized form (cortisone) to the reduced form (cortisol), and hence amplifies glucocorticoid action in metabolic tissues such as liver and adipose tissue. 11 β -HSD1 has received substantial attention as a target for the treatment of metabolic syndrome.³⁰¹ Aldosterone has a reactive aldehyde group at carbon 18 (see Fig. 12.1), which forms an 11,18-hemiacetal and is protected from dehydrogenation by 11 β -HSD2.^{4,5}

SITES OF EXPRESSION

In the kidney, 11 β -HSD2 is expressed throughout the ASDN,^{161,172,302} where it is coexpressed with MR and ENaC (see Fig. 12.9).³⁰³ It is also coexpressed in DCT,^{304,305} but consensus is building that its expression is relatively low in both DCT and early CNT, allowing cortisol access to MR.^{153,154} Interestingly, expression has also been found in the thick ascending limb,¹⁷² although at substantially lower levels. Expression is the highest in the later connecting tubule and cortical collecting duct.^{302,305} It is also expressed

in the aldosterone-sensitive segments of the colon, particularly the distal colon, as is the case for MR, although there is species variability.³⁰⁶ 11 β -HSD2 expression has also been described in several nonepithelial tissues including placenta,³⁰⁷ the nucleus tractus solitarius in the brain,³⁰⁸ and the vessel wall,⁵⁴ which makes all these tissues potential aldosterone target tissues.

IMPACT ON MINERALOCORTICOID RECEPTOR ACTIVITY

The initial^{4,5} and still widely held interpretation of the role of 11 β -HSD2 is that of excluding active glucocorticoids from the epithelial MR, which allows aldosterone unfettered access. This is only part of the picture, however; to reduce the signal-to-noise ratio from 100-fold to 10% would require that 999 of every 1000 cortisol molecules entering the cell be metabolized to cortisone, a tall order in an organ such as the kidney, which commands 20% to 25% of cardiac output. 11 β -HSD2 in epithelia (and in other tissues in which it is expressed) clearly reduces glucocorticoid levels by an order of magnitude³⁰⁹ but still leaves them with intracellular levels well above those of aldosterone. At the same time, although it is clear that when 11 β -HSD2 is operative, the glucocorticoid-occupied MR is not transcriptionally active, it is also clear that when enzyme activity is insufficient (as in apparent mineralocorticoid excess) or deficient (as in licorice abuse or by genetic mutation), cortisol can activate the MR and ion transport. Although the subcellular mechanisms involved have yet to be established, it appears that glucocorticoid-MR complexes are conformationally distinct from aldosterone-MR complexes. One intriguing possibility is that these hormone-receptor complexes, in contrast to aldosterone-MR complexes, are held inactive by the obligate generation of NADH from the cosubstrate NAD, required for the operation of 11 β -HSD2.³¹⁰ Direct evidence supports the idea that redox potential affects the activity of the glucocorticoid receptor through effects on thioredoxin.³¹¹

APPARENT MINERALOCORTICOID EXCESS: A DISEASE OF DEFECTIVE 11 β -HYDROXYSTEROID DEHYDROGENASE TYPE 2

Apparent mineralocorticoid excess was first described by New, and the molecular mechanisms responsible were established after an intense but fruitless search for a novel mineralocorticoid.³¹² The condition reflects a partial or complete deficiency of 11 β -HSD2 activity, is more common in consanguinity, and manifests as severe juvenile hypertension (see also Chapter 46).³¹³ Confectionery licorice (or that added to chewing tobacco) contains glycyrrhizic and glycyrrhetic acids, suicide substrates for 11 β -HSD2, which thus acts as a potent inhibitor of the enzyme. Lack of functional 11 β -HSD2 results in MR activation by cortisol and inappropriate mineralocorticoid-like stimulation of ENaC-mediated Na⁺ reabsorption. This causes severe hypertension, often accompanied by hypokalemia. Plasma renin, angiotensin II, and aldosterone are suppressed. Treatment of apparent mineralocorticoid excess is the

use of MR antagonists and additional antihypertensives, as required.

NONGENOMIC EFFECTS OF ALDOSTERONE

The classic effects of aldosterone on ion transport are genomic, with the MR acting at the nuclear level to regulate DNA-directed, RNA-mediated protein synthesis and thereby sodium transport. Such genomic effects are characterized by a lag period of 45 to 60 minutes before changes in ion transport can be measured, commensurate with a homeostatic role for aldosterone action in regulating sodium and potassium status in response to dietary intake. In other circumstances (e.g., orthostasis and acute blood volume depletion), aldosterone secretion rises rapidly and acute nongenomic mechanisms are needed to effect a rapid response. Such rapid effects were first demonstrated more than 30 years ago in the laboratory³¹⁴; in human vascular tissues, they have been amply demonstrated both *in vitro*³¹⁵ and *in vivo*.³¹⁶ Although most of these rapid nongenomic effects appear to be mediated via activation of the classic MR,^{314,315} there is evidence from atomic force microscopy studies for nonMR membrane sites binding aldosterone with high affinity on cultured endothelial cells.³¹⁷ Such nongenomic effects are not unique to aldosterone, having been shown for the other recognized classes of steroid hormones³¹⁸ and reported for dehydroepiandrosterone (DHEA).³¹⁹ Genomic effects commonly have a lag period and are abrogated by inhibitors of transcription, such as actinomycin D. Most nongenomic effects of steroids have time courses from onset to plateau of 5 to 10 minutes and are mediated by a variety of pathways.

Some steroid hormone receptors, such as estrogen receptors,³²⁰ are localized to the plasma membrane by myristoylation, thereby mediating rapid nongenomic signaling, but the MR does not have a myristoylation site, and there is little evidence for membrane-associated classic MR. Most rapid nongenomic effects of aldosterone do appear to be mediated by classic MRs in that they are inhibited by the specific MR antagonist, RU 28318. In some cases,³¹⁵ spironolactone is ineffective as an inhibitor. Exclusive reliance on blockade by spironolactone to define the classic MR has led to the erroneous presumption of the existence of a widely distributed aldosterone receptor that is distinct from the classic MR. However, a long search for such a membrane-bound species has been unsuccessful.³²¹ Acceptance of the nongenomic actions of aldosterone has been slow, which in part reflects the major emphasis on the clearly genomic actions of aldosterone in the kidney. Nevertheless, nongenomic effects of aldosterone are evident physiologically. The most obvious example is the conjunction of rapid secretion of aldosterone in response to orthostasis and its demonstrated rapid vascular effects.^{322,323} With more interest in the pathophysiologic effects of MR activation, particularly in nonclassic aldosterone target tissues, there has been renewed interest in the rapid nongenomic effects of aldosterone (and the physiologic glucocorticoids) via classic MR signaling. The nongenomic actions of aldosterone are discussed in further detail elsewhere.³²³⁻³²⁵

NONEPITHELIAL ACTIONS OF ALDOSTERONE

In addition to the classic epithelial tissues involved in ion transport—kidney, colon, sweat glands, and salivary glands—there are documented effects of aldosterone in the brain, vascular wall, and possibly the placenta. Many other tissues and organs have been erroneously postulated to be physiologic aldosterone target tissues, largely based on evidence that they express the MR, and can be shown *in vitro* to respond to aldosterone. What underpins these hypotheses is the misconception that aldosterone is the cognate ligand for MRs. This is true for epithelia but not for nonepithelial cells that do not express 11β -HSD2, in which cortisol is, instead, the major ligand for the MR. Cortisol was not only the ligand for the MR in cartilaginous and bony fish, millions of years before the appearance of aldosterone synthase, but is the overwhelming occupant of the MR that is not protected by 11β -HSD2 (primarily nonepithelial MR) throughout the body.

The fact that aldosterone can activate the MR under experimental conditions without 11β -HSD2 was illustrated by the work of Gómez-Sánchez and colleagues³²⁶ more than 3 decades ago. Very low doses of aldosterone, which did not affect blood pressure when infused systemically, led to elevated blood pressure when infused into the lateral ventricle of conscious, free-living rats. This did not reflect a physiologic role for aldosterone, however, as shown by the coinfusion of one, two, and five times the dose of corticosterone with aldosterone, which progressively blocked the blood pressure effect of the infused aldosterone, providing evidence for the absence of 11β -HSD2 in the hypothalamic nuclei involved and the overwhelming occupancy of the local MR by the physiologic glucocorticoid.

The two established nonepithelial aldosterone target tissues are the vascular wall and nucleus tractus solitarius in the brain. Both these tissues express 11β -HSD2, allowing aldosterone-selective MR activation; therefore both tissues can be reasonably envisaged to have important ancillary roles supporting the primary epithelial role of aldosterone on fluid and electrolyte homeostasis. Aldosterone vasoconstricts blood vessels, acutely and in the longer term, in response to volume depletion; similarly, it acts on the nucleus tractus solitarius to stimulate salt appetite. Both actions are thus harnessed into the physiologic role of aldosterone in maintaining fluid and electrolyte balance.

ACTIONS OF ALDOSTERONE ON THE VASCULATURE

Aldosterone at nanomolar concentrations in the human vascular wall causes a rapid rise in the intracellular pH, reflecting nongenomic activation of the $\text{Na}^+\text{-H}^+$ exchanger.³¹⁵ Cortisol alone over a range of doses produced no effect, but when carbenoxolone was added to inhibit 11β -HSD2, cortisol mimicked aldosterone. Inhibitor studies have revealed that the effects of both aldosterone and cortisol are mediated by a classic MR, at least in part through a nongenomic mechanism. Effects in both vascular endothelial cells and smooth muscle cells have been described.³²⁷ Of particular note, a direct role of the

smooth muscle cell MR in blood pressure regulation was demonstrated in selective gene knockout studies in mice, and its effects were shown to be at least in part due to regulation of L-type calcium channel ($\text{Cav}1.2$) expression.³²⁸ Evidence also supports a role of smooth muscle cell MR in age-related increases in vasoconstriction and rise in BP.^{329,330} In still other studies, mineralocorticoid antagonists were protective in states of tissue damage, whereas aldosterone or cortisol worsened injury. One inference from these latter results was that cortisol becomes a MR agonist in the context of tissue damage, possibly due to alteration of reactive oxygen species generation and redox potential.³³¹ To add to the complexity, aldosterone has been shown to have both vasodilatory and vasoconstricting effects in animals and humans.³¹⁹ These contradictory results have not been fully reconciled³³² but may well reflect a combination of direct stimulatory effects on vascular smooth muscle myosin light-chain phosphorylation^{332,333} on the one hand and stimulatory effects on endothelial cell nitric oxide synthase³¹⁹ on the other. Finally, it is of considerable interest that vascular smooth muscle and endothelial cells express ENaC, in addition to the MR, and that the channel might play a role in setting vascular tone and through these nonrenal mechanisms implicated in hypertension.^{334,335}

It is commonly assumed that in pathophysiologic states of high aldosterone levels, such as primary aldosteronism, the deleterious effects are mediated by aldosterone occupying and inappropriately activating nonprotected MRs in cardiomyocytes, for example. It is plausible that instead of the approximately 1% physiologic occupancy (given the ≈ 100 -fold higher levels of plasma-free cortisol), aldosterone occupancy of cardiomyocyte MRs might rise to 3% to 5%. Relatively minor degrees of MR occupancy have been shown to be effective for spironolactone, acting as a protective inverse agonist³³¹; similarly therefore minor degrees of cardiomyocyte MR occupancy by aldosterone could potentially produce the deleterious effects seen.

This explanation, however, is almost certainly incorrect. Plasma aldosterone levels are as high or higher in chronic sodium deficiency (or in the effectively volume-depleted condition of secondary hyperaldosteronism), with no deleterious cardiovascular effects. In primary and secondary aldosteronism, as well as in chronic sodium deficiency, physiologic target tissues, both renal tubular and coronary vascular, are exposed to (and respond to) maintained high levels of aldosterone. It is thus unlikely that the cardiovascular damage in primary aldosteronism reflects increased MR activation in blood vessels, coronary and peripheral. The key difference between these circumstances is that primary aldosteronism is a state of aldosterone and sodium excess and the others of sodium and volume depletion.

A plausible but untested mechanism of aldosterone-induced damage is that it is secondary to increased renal sodium reabsorption and the action of endogenous ouabain on blood vessels. Endogenous ouabain is incompletely explored, but its levels are elevated in primary aldosteronism.³³⁶ Like aldosterone, its secretion is elevated by ACTH and angiotensin (the latter via AT_2R); in stark contrast with aldosterone, it is raised (not lowered) in states of sodium excess.³³⁷⁻³³⁹ It acts via $\text{Na}^+\text{-K}^+\text{-ATPase}$ in vessel walls as a vasoconstrictor, presumably physiologically to produce a pressure natriuresis as a homeostatic response.

Thus it may be that the cardiovascular damage in primary aldosteronism reflects a combination of the effects of aldosterone plus endogenous ouabain on the vasculature; if this is the case, the source and origin of the nonepithelial effects of aldosterone remain squarely in the renal tubule and the exaggerated sodium retention therein.

TARGETING THE ALDOSTERONE PATHWAY

STEROIDAL MINERALOCORTICOID RECEPTOR ANTAGONISTS

The initial MR antagonists (MRAs), steroidal molecules, such as spironolactone, were developed shortly after the discovery of aldosterone and available for treatment of volume overload shortly thereafter.³⁴⁰⁻³⁴² The MR was then cloned in 1987³⁴³ and eplerenone was developed.³⁴⁴ Spironolactone lacks MR specificity, most notably binding with high affinity to the androgen and progesterone receptors. As well as exerting antiandrogenic and progestogenic adverse side effects (e.g., breast tenderness, gynecomastia, and decreased libido).³⁴⁵ On the other hand, while eplerenone is more specific, it is less effective in reducing blood pressure.³⁴⁶ Both agents may cause significant hyperkalemia in select populations.^{347,348} To optimize efficacy and selectivity, nonsteroidal MRAs were developed.³⁴⁹

NONSTEROIDAL MINERALOCORTICOID RECEPTOR ANTAGONISTS

Finerenone, one of several nonsteroidal MRAs, was originally derived from a dihydropyridine calcium channel blocker found to bind to and inhibit the MR.³⁵⁰ Compared with spironolactone and eplerenone, finerenone differs in tissue distribution, binding site, pharmacokinetics, and mechanism by which it inhibits the MR.^{349,351-355} Finerenone can alter the transcriptional profile of cells compared with steroidal MRAs.³⁵³⁻³⁵⁵ Finerenone is comparable in potency to spironolactone to block the MR but also has comparable specificity to eplerenone, thereby avoiding the endocrine side effects that can plague high doses or long-term use of spironolactone.³⁵⁶ Due to these differences in the pharmacology of nonsteroidal versus steroidal MRAs, agents such as finerenone have been considered a potential strategy to protect target end-organ tissues at doses that may result in significantly less hyperkalemia. Thus at equinatriuretic doses, finerenone is more potent than steroidal MRAs at protecting against end-organ injury such as cardiac hypertrophy, proteinuria, and glomerulosclerosis in preclinical and clinical studies.^{352,354,355,357-360} This is presumably because finerenone causes less MR antagonism in principal cells of the cortical collecting duct. Other nonsteroidal MRAs include esaxerenone, which has been developed and approved outside the United States for the treatment of hypertension,³⁶¹ and ocedurenone, which is similar to finerenone but has higher extrarenal tissue distribution and causes less hyperkalemia, especially in patients with advanced chronic kidney disease.³⁶²

ALDOSTERONE SYNTHASE INHIBITORS

The need for an antialdosterone therapy with more efficacy and specificity than either spironolactone or eplerenone, coinciding with the development of inhibitors of CYP11B1 for inhibition of cortisol synthesis, paved the way for the parallel discovery of inhibitors of CYP11B2 (aldosterone synthase). The aldosterone synthase inhibitor, baxdrostat, and other similar compounds (lorundrostat) are currently in clinical trials. While their clinical utility has not been fully established, aldosterone synthase inhibitors may be distinct from MRAs in that they do not block MR-mediated effects of cortisol, but may not block nongenomic actions of aldosterone.

ENAC INHIBITORS

Amiloride and triamterene are in clinical use as inhibitors of ENaC. These agents cannot inhibit ENaC-independent effects of the MR in kidney epithelial on extrarenal tissues and are primarily used to inhibit potassium secretion in states of ENaC activation, which include genetic (e.g., Liddle syndrome and gain-of-function ENaC) and acquired (licorice ingestion, posaconazole) conditions, classified as syndromes of apparent mineralocorticoid excess.

DISEASE STATES ASSOCIATED WITH ALDOSTERONE

Aldosterone is implicated in several common diseases including primary aldosteronism, congestive heart failure, chronic kidney disease, and resistant hypertension.

PRIMARY ALDOSTERONISM

Clinically, the most prevalent disorder directly involving aldosterone is Conn syndrome, or primary aldosteronism.³⁶³ In this syndrome, aldosterone secretion is elevated and (relatively) autonomous as a result of an adrenal adenoma or, more frequently, bilateral adrenal hyperplasia and, rarely, adrenal carcinoma or familial hyperaldosteronism syndromes. Once considered rare (<1% of all cases of hypertension), necessarily characterized by hypokalemia and relatively benign, primary aldosteronism is now thought to account for at least ~10% to 20% of all hypertension and a higher percentage of resistant hypertension.^{364,365} In contrast to prior teachings, frank hypokalemia is found in only 25% to 30% of cases, and the incidence of cardiovascular pathology (e.g., fibrosis, fibrillation, infarct, and stroke) is substantially higher than in age-, gender-, and blood pressure-matched individuals with essential hypertension.³⁶⁶ For further discussion of its role in clinical hypertension, see [Chapter 46](#).

Guidelines for the case detection, diagnosis, and management of primary aldosteronism have been published³⁶⁷ as a first step in addressing what has been increasingly recognized as an important public health issue. It has long been thought and taught that the role of aldosterone in blood pressure regulation reflects its epithelial

effects leading to retention of sodium and with it water, which thus increases circulating volume. This increase, in turn, is reflected in an increased cardiac output, which is reflexively normalized by vasoconstriction and thus elevation of blood pressure (in keeping with the Guyton hypothesis³⁶⁸). Although the epithelial effects of aldosterone on vascular volume are indisputably homeostatically important, there have been compelling experimental and clinical studies to suggest a role for nonepithelial effects in mineralocorticoid-induced hypertension.^{369,370} In addition to MR-mediated central nervous system and vascular effects in hypertension, roles for macrophages have been demonstrated.^{371,372}

Primary hyperaldosteronism may be caused by gain-of-function mutations in genes that regulate aldosterone synthesis in the zona glomerulosa. These findings have emerged from studies of somatic mutations in aldosterone-producing adenomas and germ-line mutations in rare forms of familial hyperaldosteronism. More than 90% of aldosterone-producing adenomas carry somatic mutations in K⁺ channel Kir3.4 (*KCNJ5*), Ca²⁺ channel Ca_v1.3 (*CACNA1D*), α -1 subunit of the Na⁺/K⁺ ATPase (*ATP1A1*), plasma membrane Ca²⁺ transporting ATPase 3 (*ATP2B3*), Ca²⁺ channel Ca_v3.2 (*CACNA1H*), Cl⁻ channel CLC-2 (*CLCN2*), β -catenin (*CTNNB1*), and/or G-protein subunits alpha q/11 (*GNAQ11*). Several of these genes, as well as the gene associated with glucocorticoid-remediable hyperaldosteronism (*CYP11B1/2*), are implicated in familial forms of hyperaldosteronism.^{25,26,373} See the earlier section on "Aldosterone Synthesis" for additional details. Treatment of primary aldosteronism may include steroidal MRAs or adrenalectomy for unilateral aldosterone secretion and steroidal MRAs for bilateral aldosterone secretion. For patients eligible for adrenalectomy, clinical studies comparing these treatment modalities for prevention of major adverse cardiovascular or kidney events are lacking. Nonsteroidal MRAs, aldosterone synthase inhibitors, and ENaC inhibitors are of potential benefit in patients with primary aldosteronism who cannot tolerate or are inadequately treated with steroidal MRAs. However, phase 3 noninferiority studies are lacking.

CONGESTIVE HEART FAILURE

Aldosterone has been implicated in the pathophysiology of congestive heart failure since its discovery in the mid-1950s.^{374,375} Until fairly recently, most of the focus was on the counterproductive effects of aldosterone in epithelia, such as salt and fluid retention. More recently, the beneficial effects of MRAs in congestive heart failure have suggested an additional effect on myocardium itself.³⁷⁶ In the Randomized Aldactone Evaluation Study (RALES),³⁷⁶ addition of low-dose (mean, 26 mg/day) spironolactone to standard-of-care treatment in patients with progressive heart failure produced a 30% reduction in mortality and 35% fewer hospitalizations. This result is often attributed to spironolactone inhibition of the effect of aldosterone on cardiomyocyte MRs but actually may reflect inhibition of cortisol, which acts as a MR agonist under ischemic conditions. Subsequently, the Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study (EPHESUS) examined the effect of the MR antagonist eplerenone, with improved specificity relative to spironolactone, on heart failure due to systolic

dysfunction complicating acute myocardial infarction. The study showed that adding eplerenone (25–50 mg/day) to conventional therapy significantly decreased all-cause and cardiovascular mortality.³⁷⁷ Potassium levels were only slightly higher in the eplerenone-treated group than in the placebo-treated group (4.47 mmol/L and 4.54 mmol/L, respectively). Coupled with studies on the direct vascular effects of aldosterone discussed earlier, these data suggest that MR antagonists have a beneficial effect that cannot be accounted for by diuretic actions in the kidney alone.³⁷⁸

It is also notable that a trial (Eplerenone in Mild Patients Hospitalization And Survival Study in Heart Failure [EMPHASIS-HF]) examining the effect of eplerenone in New York Heart Association (NYHA) class II heart failure (milder than previously examined) was stopped early because significant benefit was found in the treated group.³⁷⁹

Both spironolactone and eplerenone cause high rates of hyperkalemia.^{347,348} At-risk populations include patients with type 2 diabetes and/or CKD.³⁸⁰ Thus a nonsteroidal MRA with less effect on ENaC in the ASDN, but with comparable or distinct inhibition of the MR in cardiomyocytes, may be useful for reducing major adverse cardiovascular events in these cohorts. In the Finerenone in Reducing Cardiovascular Mortality and Morbidity in Diabetic Kidney Disease (FIGARO-DKD) trial, eligible patients had type 2 diabetes mellitus and mild to moderate albuminuria and stage 2 to 4 CKD or macroalbuminuria and milder CKD. Patients were similar to the RALES or EPHESUS trials in that patients were already taking angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers. During a median follow-up of 3.4 years, finerenone versus placebo reduced primary outcome events (hazard ratio, 0.87; 95% confidence interval [CI], 0.76 to 0.98; $P = 0.03$), with a significantly higher rate of hyperkalemia-induced discontinuation compared with placebo but a historically lower incidence of hyperkalemia-related discontinuation compared with steroidal MRAs.³⁸¹ The American Diabetes Association has incorporated nonsteroidal MRAs into their guidelines and recommended use of finerenone for similar indications.³⁸²

CHRONIC KIDNEY DISEASE

The role of MR blockade in slowing the progression of CKD has been considered. However, the risk of hyperkalemia with steroidal MRAs has precluded major, randomized placebo-controlled trials to slow progression of CKD and/or decrease major adverse kidney events. To assess the role of a nonsteroidal MRA, finerenone was compared with placebo in the Finerenone in Reducing Kidney Failure and Disease Progression in Diabetic Kidney Disease (FIDELIO-DKD) trial.³⁸³

This study was similar to the FIGARO-DKD trial in its inclusion of patients with type 2 diabetes mellitus and mild to moderate albuminuria with stage 2 to 4 CKD or macroalbuminuria and milder CKD, but it was designed to evaluate renal endpoints instead of cardiovascular endpoints. All the patients were treated with renin-angiotensin system blockade. The primary composite outcome, assessed in a time-to-event analysis, was kidney failure, a sustained decrease of at least 40% in the eGFR from baseline, or death from renal causes. During a median follow-up of

2.6 years, finerenone demonstrated an 18% lower risk of CKD progression and cardiovascular events (hazard ratio, 0.82; 95% confidence interval [CI], 0.73–0.93; $P = 0.001$). Similar to the FIGARO-DKD trial, patients on finerenone had a significantly higher rate of hyperkalemia-induced discontinuation compared with placebo but a historically lower incidence of hyperkalemia-related discontinuation compared with steroidal MRAs.

RESISTANT HYPERTENSION

Resistant hypertension is associated with heightened risks of death, major cardiovascular events, left ventricular hypertrophy, progressive kidney disease,³⁸⁴ and cognitive impairment compared with patients with less severe forms of hypertension and/or normotensive controls.^{385–390} PATHWAY-2 showed that spironolactone was a superior agent to lower blood pressure in resistant hypertension compared with placebo, doxazosin, or bisoprolol.

Baxdrostat, an aldosterone synthase inhibitor, has also shown promising results in a phase 2 study of patients with resistant hypertension.³⁹¹ Compared with placebo, the difference in the change in systolic blood pressure between the 2-mg group and the placebo group was -11.0 mm Hg (95% confidence interval [CI], -16.4 to -5.5 ; $P < 0.001$, $N = 248$ patients). A phase 3 study is needed to confirm this, but aldosterone synthase inhibitors may provide similar protection to steroidal MRAs.

A companion study of PATHWAY-2 compared the response of spironolactone to high-dose amiloride to demonstrate similar efficacy for lowering blood pressure ($r^2 = 0.64$, $P < 0.0001$). These data are consistent with an effect on ENaC inhibition as an important mechanism in resistant hypertension.³⁹²

CONCLUSION

In summary, the pathophysiologic effects of aldosterone excess in primary aldosteronism have been well documented, and the MR plays an important role in heart failure, CKD, and hypertension. Importantly, the MR expressed in nonepithelial cells may commonly be activated by cortisol rather than aldosterone, which is present in serum at levels that are higher by 100-fold or more than aldosterone and mimics aldosterone in the context of tissue damage.

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QUESTIONS

1. The most appropriate initial treatment for a patient with primary aldosteronism due to bilateral adrenal hyperplasia is:
 - a. Triamterene
 - b. Bilateral adrenalectomy
 - c. Spironolactone
 - d. β -blockers
 - e. Thiazide diuretics

Answer: c

Rationale: In bilateral adrenal hyperplasia, there is no discrete aldosterone producing adenoma that can be removed. Removal of both adrenals results in hypoadrenalism with lifelong tendency toward Addisonian crisis. Moreover, the results are unpredictable, and spironolactone and other mineralocorticoid receptor antagonists are effective.

2. The enzyme conferring aldosterone specificity on mineralocorticoid receptors in epithelial tissues is:
 - a. CYP11B1
 - b. CYP11B2
 - c. 11β HSD1
 - d. 11β HSD2
 - e. Aldosterone synthase

Answer: d

Rationale: 11β HSD2 converts cortisol (and corticosterone) to an inactive metabolite leaving MR free to bind aldosterone.

3. Mineralocorticoid receptors are aldosterone regulated:
 - a. Transcription factors.
 - b. Receptor tyrosine kinases.
 - c. G-protein coupled receptors.
 - d. Mitochondrial enzymes.
 - e. Epithelial sodium channels.

Answer: a

Rationale: It is well established that MR bind DNA and regulate gene transcription when activated by aldosterone (or cortisol).

4. Serum- and glucocorticoid-regulated kinase 1 (SGK1) acts in principal cells to:
 - a. Inhibit K^+ channels
 - b. Stimulate the epithelial sodium channel (ENaC)
 - c. Stimulate aldosterone uptake
 - d. Stimulate cortisol break down
 - e. Inhibit the Krebs cycle

Answer: b

Rationale: A broad range of animal and cell culture experiments support the conclusion that the main action of SGK1 is to stimulate ENaC.

5. Which of the following is true?
 - a. Acute aldosterone-regulated potassium secretion is primarily via direct stimulation of potassium channels.
 - b. Mineralocorticoid receptors are exclusively expressed in the distal nephron.
 - c. Hyperkalemia-induced aldosterone secretion leads to high blood pressure.
 - d. Sodium reabsorption increases the driving force for potassium and proton efflux in the distal nephron.

Answer: d

Rationale: The favored answer is d, however, give yourself partial credit for “c”: recent evidence supports the idea that very high K^+ intake can increase blood pressure